Temperature-dependent tensile and fatigue behaviors of additively manufactured A6061-RAM2 aluminum alloy: A comparative analysis between L-PBF and LP-DED

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Highlights

● Temperature-dependent mechanical behavior of A6061-RAM2 manufactured using L-PBF and LP-DED A6061-RAM2 was analyzed

● L-PBF specimens exhibited finer grains which resulted in higher strength up to 200 °C compared tothe LP-DED specimens

● Slower crack growth in L-PBF specimens at high strain level gave longer fatigue life

● LP-DED specimens had shorter life at low strain level due to larger carbide particles

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Abstract

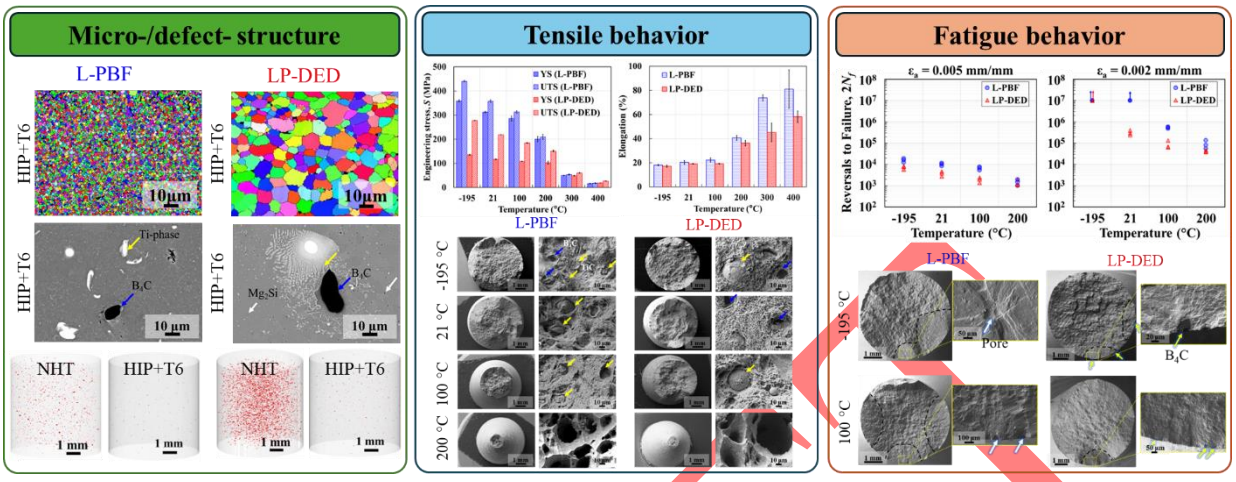
This study investigated the tensile and fatigue behaviors of laser powder bed fused (L-PBF) and laser powder direct energy deposited (LP-DED) A6061-RAM2 aluminum alloy across a wide temperature range. Tensile and strain-controlled fatigue tests (fully reversed at 0.005 and 0.002 mm/mm amplitudes) were conducted from cryogenic (-195 °C) to elevated temperatures (up to 400 °C for tensile and 200 °C for fatigue). Tensile strengths for both processes gradually declined from -195 to 200 °C due to a gradual increase in dislocation mobility, with finer-grained L-PBF specimens exhibiting higher strengths compared to coarser-grained LP-DED specimens. A sharp strength drop occurred between 200 and 400 °C due to excessive grain boundary sliding, yielding comparable strengths for both processes. For fatigue behavior, at the higher strain amplitude of 0.005 mm/mm, L-PBF specimens showed longer fatigue lives than LP-DED ones from -195 to 100 °C due to slower crack propagation. However, at the lower strain amplitude of 0.002 mm/mm, LP-DED specimens exhibited shorter fatigue lives than L-PBF ones from 21 to 200 °C, due to crack initiation from larger B_4C particles in LP-DED compared to smaller volumetric defects in L-PBF.

Keywords

Laser powder bed fusion (L-PBF/LB-PBF); Laser powder direct energy deposition (L-DED/LP-DED); A6061-RAM2; Tensile and fatigue behaviors; Cryogenic and elevated temperatures

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Graphical abstract



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Abbreviations and nomenclature

AM	Additive manufacturing/additively manufactured
BSE	Backscattered electron
EBSD	Electron backscatter diffraction
EDS	Energy-dispersive X-ray spectroscopy
EL	Elongation to failure
GB	Grain boundary/grain boundaries
HIP	Hot isostatic pressing/hot isostatically pressed
HT	Heat treatment/heat-treated/ heat treatments
IPF	Inverse pole figure
L-PBF	Laser powder bed fusion/fused
LP-DED	Laser powder direct energy deposition/deposited
MMC	Metal matrix composite
NHT	Non-heat treated
RAM	Reactive additive manufacturing
SEM	Scanning electron microscope
UTS	Ultimate tensile strength
X-CT	X-ray computed tomography
YS	Yield strength
$2N_f$	Reversals to failure
D_{10}	10th percentile of powder particle size
D_{50}	50th percentile of powder particle size
D_{90}	90th percentile of powder particle size
ε_a	Strain amplitude
ε_{max}	Maximum strain
ε_{min}	Minimum strain
e	Engineering strain
L_0	Original gage length of tensile specimen
L	Final gage length of tensile specimen
R_ε	Ratio of minimum strain to maximum strain
S	Engineering stress
σ_a	Stress amplitude
σ_m	Mean stress
T_m	Melting temperature in Kelvin
√area	Square root of projected area of a defect on the loading plane

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1. Introduction

A6061 is a heat-treatable, precipitation-hardened Al-Si-Mg alloy widely used across numerous applications in aerospace, automotive, marine, and other sectors due to its combination of good strength-to-weight ratio, corrosion resistance andmachinability. These applications include aircraft wings, spacecraft fuel tanks, automotive engine cylinder blocks, and marine structures, which often encounter environments of wide temperature ranges from cryogenic to elevated [1–4]. Components in these applications can have a combination of complex geometries (e.g., thin walls, mesh structures, and internal features), which can be difficult to manufacture using conventional processes. Additive manufacturing (AM) with its near-net shaping ability and reduced lead times and costs offers potential to address these challenges [5,6]. However, tA6061 has poor flowability in the molten state and high solidification shrinkage, which can lead to hot tearing and the formation of volumetric defects/solidification cracks during AM process [7,8]. Additionally, its relatively high thermal conductivity also contributes to microstructural and mechanical anisotropy in the fabricated parts [9,10].

Reactive additive manufacturing (RAM), a grain refinement technique patented by Elementum 3D, offers an approach to overcome these shortcomings [11–13]. For the RAM of A6061, a small amount of inoculant particles, such as Ti and B_4C, are blended with pre-alloyed A6061 powder feedstock, to promote heterogeneous nucleation during solidification. The abundance of fine solid nuclei disrupts the formation of dendrites and thereby improves the flow of the remaining liquid phase, helps distribute thermal stresses more evenly, prevents solidification cracking, and alleviates the formation of volumetric defects [8]. The inoculant particles can also serve as grain boundary (GB) pinners to resist grain growth, preserving the heterogeneous solidification's beneficial effect to produce an isotropic, fine-grained, and equiaxed

microstructure. The product of RAM is typically a dispersion-strengthened metal matrix composite (MMC) [14]. The RAM alloys are typically designated with a suffix "-RAMX", where X stands for the volume fraction of the inoculant particles. For A6061, the most common RAM form is the A6061-RAM2 indicating 2% inoculant nanoparticles.

Before deploying additively manufactured (AM) alloys, their characteristics in microstructure and defect content and the resulting mechanical behavior pertaining to their service temperature range must be carefully understood. Each AM process used to build the alloy is also critical, since the potential variations in processing conditions among different AM processes can give rise to different thermal histories and result in different micro-/defect-structures. Although AM processes share a few common traits, such as high solidification or cooling rates, repeated melting or heating, and highly dynamic melt pools, they may vary significantly from one process to another [15]. For instance, the cooling rates of laser powder bed fusion (L-PBF) and laser powder directed energy deposition (LP-DED), two of the most commonly used AM processes [16–19], range from 10^5 to 10^7 K/s [20] and from 10^2 to 10^5 K/s [21], respectively. This can largely be ascribed to LP-DED employing larger spot size (an order of magnitude larger), higher powered lasers (2-5 times more power) and lower scan velocities (1-2 orders of magnitude lower), which produce much wider and deeper melt pools than L-PBF [22,23]. This results in L-PBF produced material with finer microstructures and lack of fusion defects and gas entrapped pores [24,25] and while LP-DED tends to induce coarser microstructures and keyhole defects [26,27].

In general, the differences in micro-/defect-structure between these two processes can greatly influence the mechanical, both tensile and fatigue, behavior of the alloy at different temperatures. At room temperature, the coarser grain structures in LP-DED typically result in lower strength [28]. However, at elevated temperatures, the finer grain structures in L-PBF become more

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susceptible to GB sliding, which can significantly reduce the strength [29]. The different defect contents and microstructures of the parts fabricated by the two processes can respectively result in distinct high-cycle and low-cycle fatigue behaviors, where crack initiation and crack growth play critical roles [30]. For instance, at room temperature and in the high-cycle fatigue regime, the volumetric defects and their features in the fine-grained microstructures of L-PBF alloys can dictate the crack initiation and thus the fatigue life. In the low-cycle fatigue regime, slower crack growth is often observed for coarse-grained microstructures, such as those produced by LP-DED, due to the larger plastic zone ahead of crack tips, increased energy dissipation, and the resulting plasticity-induced crack closure [31–34]. Such influences may also depend on the test temperature [35]. For instance, at elevated temperatures, enhanced plasticity can promote crack initiation from microstructural features, making materials less sensitive to defects. The excessive GB sliding occurring in fine-grained microstructures for L-PBF at such temperatures can accelerate crack propagation [36,37].

The characterization of the A6061-RAM2 alloy's microstructure, defect content in L-PBF and LP-DED, and their effects on tensile and fatigue behaviors across a wide temperature range is currently incomplete. The existing literature has primarily focused on the printability, microstructure, and room temperature tensile properties of the A6061-RAM2 alloy produced by the L-PBF process. The RAM technique results in the formation of fine-grained and equiaxed microstructures, which improve the alloy's tensile strength, due to grain boundary strengthening [38–41]. However, the microstructure of the A6061-RAM2 alloy produced using LP-DED has not been researched yet. Additionally, the role of volumetric defects on the fatigue behavior of the A6061-RAM2 alloy fabricated by any AM technique has not been studied.

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This study investigated the tensile and fatigue behaviors across different test temperatures of the AM A6061-RAM2 alloy fabricated via L-PBF and LP-DED techniques. The uniaxial tensile behaviors of the alloy were assessed at -195, 21, 100, 200, 300, and 400 °C. Additionally, the fully reversed strain-controlled fatigue behavior was examined at two different strain amplitudes of 0.005 and 0.002 mm/mm and at -195, 21, 100, and 200 °C. The failure mechanisms of the alloy for both processes under tensile and fatigue loading are analyzed and discussed.

2. Experimental procedures

2.1. Material and specimen fabrication

The A6061-RAM2 powders, supplied by Elementum 3D and used for L-PBF and LP-DED processes, were prepared throughvacuum induction gas atomization of pre-alloyed A6061 and subsequent mechanical blending with RAM additives.. The chemical compositions of the powders for both processes are listed in **Table 1**, which have been measured by the powder provider via inductively coupled plasma optical emission spectrometry (ICP-OES), except for O, N, and C contents. The O and N contents were determined using the inert gas fusion according to ASTM E1409-13 [42], and the C content was measured using the combustion infrared absorption method following ASTM E1019-18 [43]. Scanning electron microscopy (SEM) images of the particles of both powder batches are presented in **Figure S1** in the Supplemental Material. The size range of powder particles was 15-63 µm for L-PBF and 53- 120 µm for LP-DED. The powder particle size distribution parameters (e.g., the 10th, 50th, and 90th percentiles (D_{10}, D_{50}, and D_{90})), determined by laser diffraction particle size analysis, are presented in

Table 2.

Table 1. The chemical composition (wt.%) of A6061-RAM2 powder batches used for L-PBF and LP-DED processes.

Composition (wt.%)		
Elements	**L-PBF**	**LP-DED**
Al	Bal.	Bal.
B	0.72	0.78
C	0.23	0.22
Cu	0.26	0.26
Mg	0.89	0.93
Ti	3.04	2.10
Si	0.66	0.55
Cr	0.09	0.05
Fe	0.15	0.06
Ni	0.01	0.01
V	0.01	0.01
O	0.04	0.015
N	<0.01	<0.01
Zn	0.01	0.01
Other each	<0.05	<0.05
Other total	<0.15	<0.15

Table 2. The powder particle size distribution parameters for L-PBF and LP-DED processes.

Process	D_{10} (µm)	D_{50} (µm)	D_{90} (µm)
L-PBF	15.70	32.20	57.90

LP-DED	56.95	79.23	119.40

The L-PBF specimens were manufactured using an EOS M290 machine. A stripe scanning strategy with a scan direction rotation of 67° between layers was applied, and the substrate was pre-heated to 80 °C. The LP-DED specimens were fabricated by an RPM Innovations (RPMI) 557 laser deposition system. A coaxial nozzle delivered the powder into the molten pool. Cylindrical bars with dimensions of 14 mm in diameter and 100 mm in length were fabricated vertically using both processes with the process parameters listed in **Table 3**. The fabrication processes were performed in a fully inert Ar atmosphere to prevent oxidation.

Table 3. The process parameters used in L-PBF and LP-DED machines to fabricate cylindrical bars of A6061-RAM2 alloy.

Process	Laser power (W)	Scan speed (mm/s)	Hatch distance (mm)	Layer thickness (µm)	Powder feed rate (g/min)
L-PBF	370	1400	0.2	30	–*
LP-DED	1500	12.7	N/A**	N/A**	2.7

* This process parameter does not apply to L-PBF.
** N/A stands for data not available, since the manufacturer of the LP-DED specimens, RPMI, did not share the data.

2.2. Heat treatment

After fabrication, the cylinder bars underwent multiple HT steps, as shown in **Figure 1**. These steps followed AMS 2770 [44], which specifies solutionizing and aging parameters; and ASTM F3301-18a [45], which specifies hot isostatic pressing (HIP) parameters. The HT steps included HIP followed by T6 treatment. HIP was conducted at 510 °C under 100 MPa isostatic pressure for 2 hrs followed by furnace cooling to room temperature. The T6 treatment involved

solutionizing at 530 °C for 2 hrs followed by quenching to room temperature, and finally aging at 160 °C for 18 hrs followed by air cooling. Glycol and Ar were used as quenching media for L-PBF and LP-DED bars, respectively.

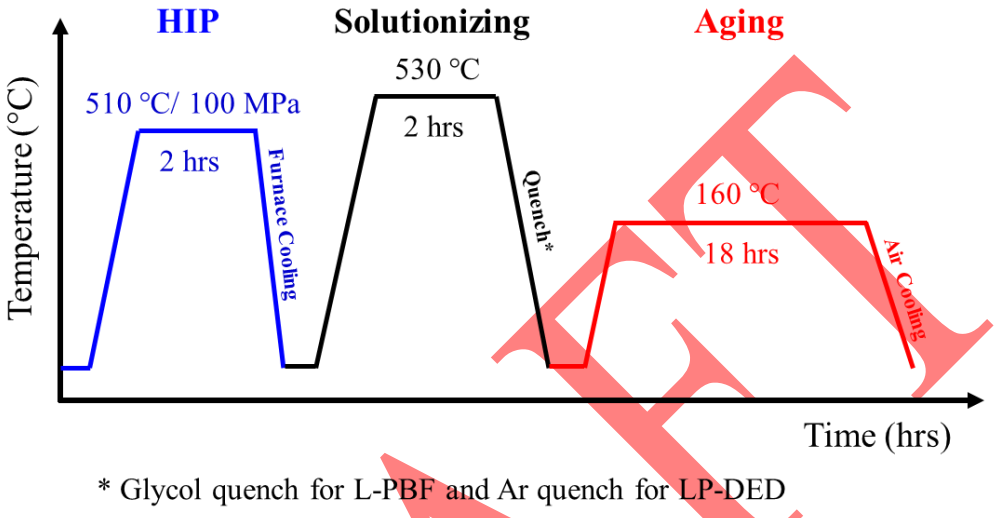


Figure 1. A schematic representation of the HT schedules used for this study for both L-PBF and LP-DED A6061-RAM2 cylindrical bars.

2.3. Microstructure and defect characterizations

The microstructure characterization was completed on coupons excised from the cylindrical bars on the longitudinal plane (i.e., parallel to the build direction). For each process, one coupon was analyzed before the HT, labeled as non-heat treated (NHT); and the other was after the full HT, labeled as HIP+T6. The coupons were mounted using epoxy resin and then ground and polished according to ASTM E3 [46]. The coupons were ground using silicon carbide (SiC) sandpapers ranging from 320 to 4000 grits under a compressive force of 5N. Subsequently, the surfaces were further polished using Buehler ChemoMet and colloidal silica of 0.05 µm particle size. Microscopy was conducted at a voltage of 10kV and a current of 1.5 nA using a Zeiss Crossbeam 550 Scanning Electron Microscope (SEM) equipped with electron backscatter

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diffraction (EBSD) and energy dispersive X-ray spectroscopy (EDS) detectors. Backscattered secondary electron (BSE) images were obtained using the electron channeling contrast imaging technique.

The X-ray computed tomography (X-CT) was conducted on 5 mm diameter and 5 mm height coupons, which were machined from the mid-section of cylindrical bars for each fabrication process. The same coupon from each process was examined by X-CT in the NHT condition, after HIP, and after T6 to evaluate the evolution of volumetric defects during the HT process. Scans were performed using a Zeiss Xradia 620 Versa system, with a 0.4X objective lens, 60 kV accelerating voltage, 6.5 W power, and 5.3 µm voxel size. Volumetric reconstruction was done using Zeiss proprietary software, followed by image analysis using ImageJ and Dragonfly software.

2.4. Mechanical testing and fractography

Out of fifty fully heat-treated (HT) cylindrical bars from both L-PBF and LP-DED processes, twenty were machined into tensile specimens following ASTM E8 [47] (see **Figure 2** (a)). The uniaxial tensile tests were conducted at -195, 21, 100, 200, 300, and 400 °C, with three specimens tested at each temperature. The room temperature (21 °C) tensile tests were carried out according to ASTM E8 [47], while cryogenic (-195 °C) and elevated temperature (100, 200, 300, and 400 °C) tests were performed according to ASTM E1450 [48] and ASTM E21 [49], respectively. For cryogenic tests, specimens were immersed in liquid nitrogen until the temperature reached -195 °C. They were soaked for 5 min and then pulled to failure. After the failure, the fractured specimens were removed from liquid nitrogen and left in the open air. For elevated temperature tests, a clam-shell radiant heat furnace was used. Two thermocouples were attached to the specimen in each test

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to monitor the temperature. The specimens were heated to the test temperature and soaked for 30 min before being pulled to failure. After the failure, the fractured specimens were promptly quenched in water. All the specimens were tested at a constant strain rate of 0.005 mm/mm/min. An extensometer was attached to the gage section to measure the axial strain of the specimens until the strain reached 0.1 mm/mm. After that, the extensometer was removed, and then the test was continued in displacement control until fracture. The final gage length, L, was measured by fitting the two halves of the broken specimen together. The specimen's elongation to failure (EL) was calculated using EL = $(L - L_0) / L_0 \times 100\%$, where L_0 is the initial gage length. Each tensile property, i.e., yield strength (YS), ultimate tensile strength (UTS), and EL, represents the average of three tests performed for each temperature.

The remaining thirty HT cylindrical bars from each process were machined into fatigue specimens, as shown in **Figure 2** (b), following ASTM E606 [50]. All machined specimens were polished along the loading direction to ensure smooth surfaces. The uniaxial fully reversed ($R_\varepsilon = \varepsilon_{min}/\varepsilon_{max} = -1$) fatigue tests were performed in a strain-controlled mode according to ASTM E606 [50] at the temperatures of -195, 21, 100, and 200 °C. These tests were conducted at two different strain amplitudes of 0.005 mm/mm and 0.002 mm/mm using a triangular waveform. The loading frequencies were 0.5 Hz and 1.25 Hz, respectively, to maintain a nominal cyclic strain rate of 0.010 mm/mm/sec. Before testing, specimens were isothermally held at the test temperatures, with a soaking time of approximately 5 minutes for cryogenic (-195 °C) tests and 30 minutes for elevated temperature tests (100 and 200 °C), which were consistent with the tensile tests. Three tests were performed for each temperature and strain amplitude combination. Tests that reached 10^7 reversals ($2N_f$) were considered as runout.

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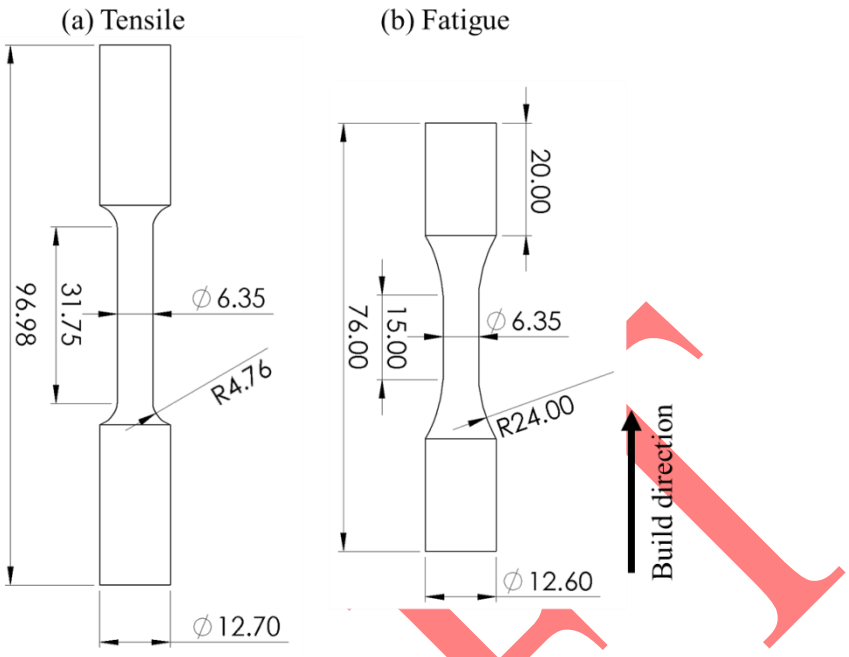


Figure 2. The geometry of (a) tensile and (b) fatigue test specimens following ASTM E8 [47], and ASTM E606 [50], respectively. Note that all the dimensions are in 'mm'.

3. Effect of manufacturing processes

3.1. Micro-/defect-structure

The X-CT analysis results showing the defect contents of the same coupon from each process evolving through NHT, HIP, and HIP+T6 conditions are presented in **Figure 3** (a)-(c). The quantitative analysis of defects for both processes is presented using the histograms in the **Figure 3** (d). To avoid false detection, defects with an equivalent diameter smaller than three voxels (~20 µm) were not reported in both 3D visualizations and the histograms.

The defect population for the NHT LP-DED condition (relative porosity of 0.04%) was larger than the NHT L-PBF condition (relative porosity of 0.02%) at all size intervals. The maximum defect size, measured by the equivalent sphere diameter, was 97 µm for the L-PBF coupon and 106 µm for the LP-DED one. After HIP, no defects were detected in coupon from

either process. However, after HIP+T6 treatment, a few defects such as gas entrapped pores reappeared in the coupons. The reappearance of these pores can be attributed to the presence of trapped H_2 gas within the material [51,52]. During fabrication, under high laser energy as the powder was melted, atomic H formed primarily due to the presence of moisture (H_2O) in the powder and its reaction with metallic particles [51]. As the metal solidified, atomic H was entrapped as gas pores [53]. During HIP, under high pressure and temperature, the H trapping pores shrank and became too small to detect with the 5.3 μm voxel size used for X-CT scanning. In the subsequent T6 treatment, conducted without pressure, the trapped gas pores expanded and reappeared as small detectable pores [54].

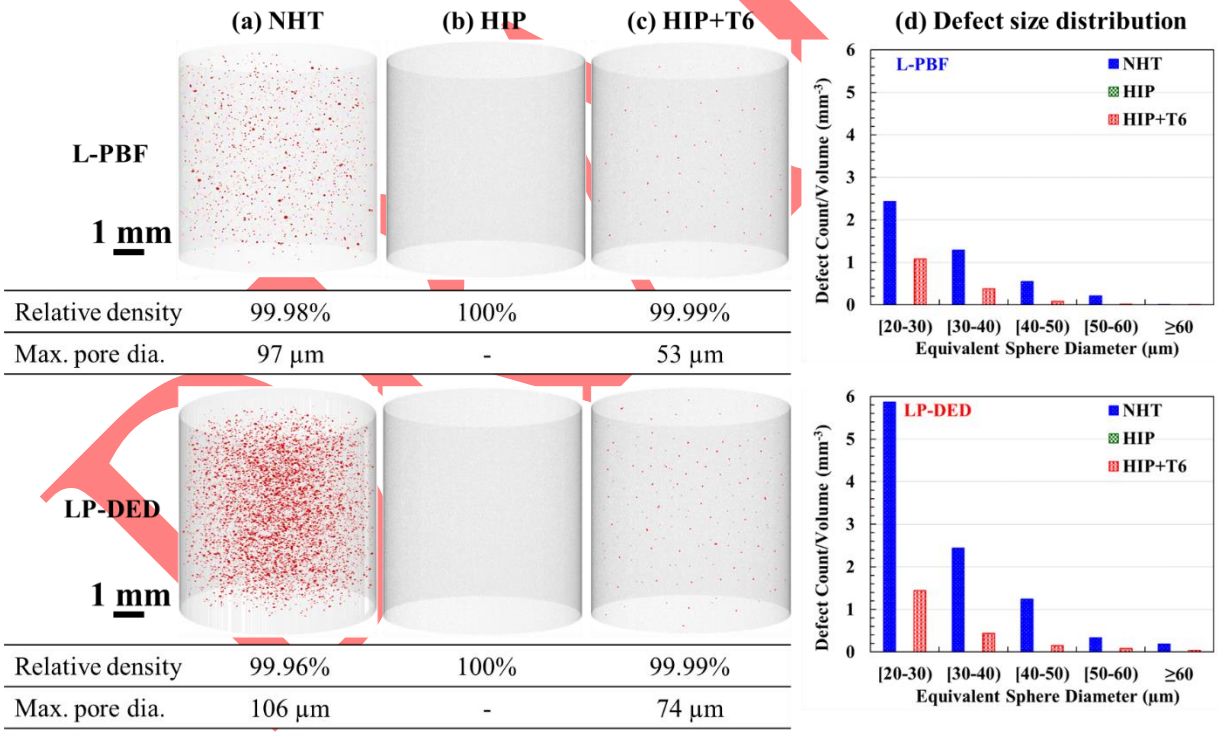


Figure 3. X-CT scan results of coupons for L-PBF and LP-DED A6061-RAM2 alloy, scanned sequentially in NHT, HIP, and HIP+T6 conditions, showing (a)-(c) 3D visualization of volumetric defects and (d) the corresponding defect size distributions.

The BSE micrographs of the A6061-RAM2 coupons on the longitudinal plane (i.e., the plane parallel to the build direction) in NHT and HIP+T6 conditions for both L-PBF and LP-DED processes are presented in **Figure 4**. At a lower magnification (see **Figure 4** (a)-(b)), the L-PBF microstructure in the NHT condition contained more secondary phase particles, such as B_4C and unreacted or partially reacted TiC particles than the LP-DED microstructure. At a higher magnification (see **Figure 4** (c)-(d)), notable differences in size and shape among these particles became evident.

The B_4C particles, appearing in dark contrast in the micrographs for LP-DED, were irregular and larger compared to those for L-PBF ones. Additionally, TiC particles, appearing in bright contrast, displayed random shapes and were smaller in the L-PBF microstructure than those in the LP-DED one, where they were circular and larger. These size differences can be attributed to the different thermal histories of the two processes [22,23]. In LP-DED the higher laser power and slower cooling of the melt pools kept the material molten for a longer time at higher temperatures. This longer time at high temperatures allowed for particle coarsening during the molten state and after solidification due to the slower cooling rate. This is in line with the findings of Chu et al. who found that in an Al/TiC MMC system TiC particles coarsened from 2 µm to 10 µm when exposed to higher temperatures for longer times in the molten state from 5 to 30 minutes [55].

After the HIP+T6 HT, B_4C and TiC particles remained in the microstructures for both processes (see **Figure 4** (g)-(l)). The EDS elemental maps in **Figure 5** (a)-(c) for the fully HT L-PBF and LP-DED microstructures confirm the presence of these particles. Additionally, some TiC particles were surrounded by a reaction layer composed of Al_3Ti and Ti_5Si_3, which are the phases also identified in various A6061 MMC systems in the literature [41,56]. Mg_2Si strengthening

phases are typically expected to precipitate in the A6061 alloy after the T6 HT. The BSE micrographs at a higher magnification revealed the presence of Mg_2Si particles along the grain boundaries (GB) in both microstructures after HT (see **Figure 4** (k)-(l)) and the presence of these particles can be confirmed by the EDS elemental maps (see **Figure 5** (b) and (d)).

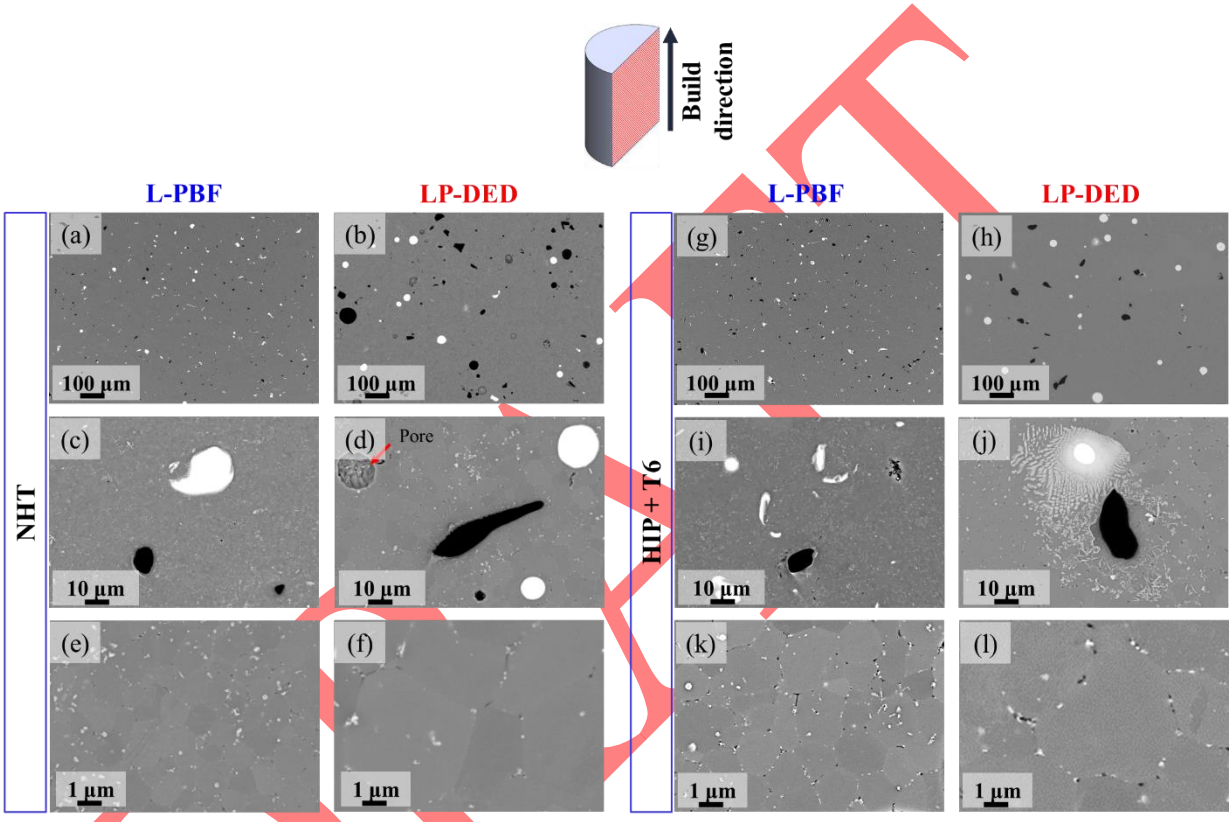


Figure 4. The BSE images for both L-PBF and LP-DED A6061-RAM2 alloy in different heat treatment conditions: (a)-(f) NHT, and (g)-(l) HIP+T6. The red arrow indicates a pore.

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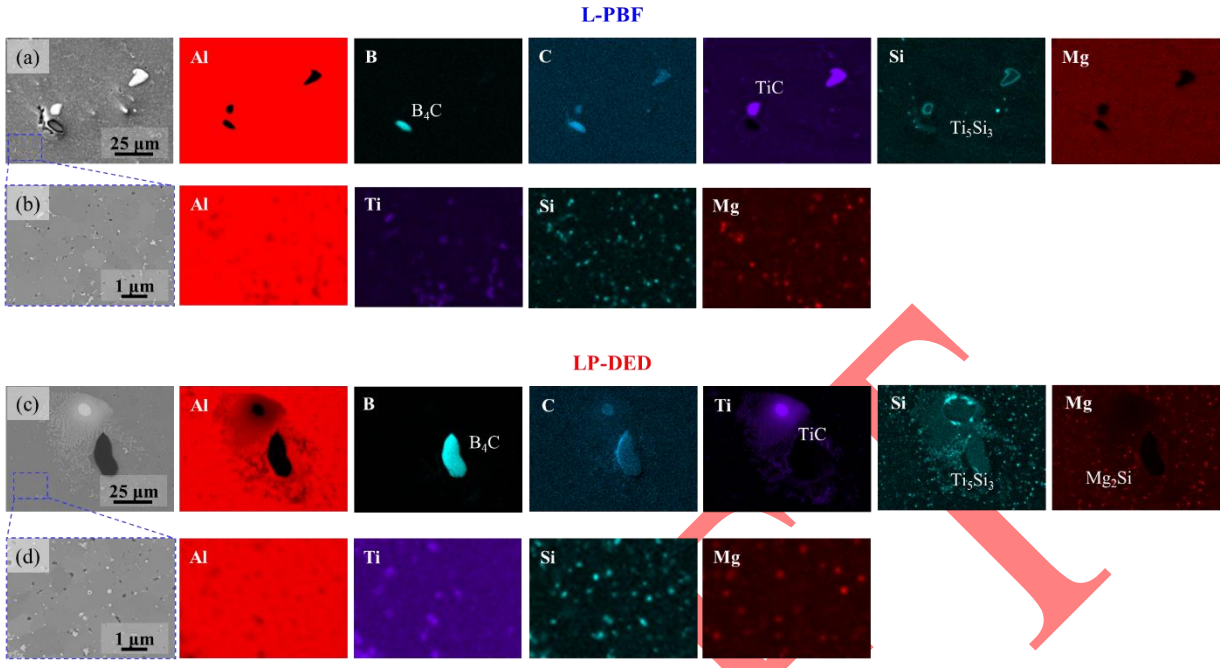


Figure 5. Results from EDS analysis, the elemental maps for (a)-(b) L-PBF, and (c)-(d) LP-DED A6061-RAM2 in fully HT condition.

The longitudinal plane grain structures of L-PBF and LP-DED A6061-RAM2 coupons in NHT and HIP+T6 conditions are shown in **Figure 6**. Inverse pole figure (IPF) maps from the EBSD analysis confirmed the presence of equiaxed grain structures in the coupons, in the NHT condition (see **Figure 6** (a)-(b)). The TiC and B_4C inoculant particles provided numerous heterogeneous nucleation sites for solidification, promoting grain refinement and the formation of equiaxed grains. The more secondary phase particles in the L-PBF microstructure (see **Figure 4** (a)-(b)) provided more nucleation sites and resulted in a finer grain structure. The L-PBF microstructure is comprised of significantly finer grains, with an average equivalent circle diameter of around 1.5 µm, compared to around 6 µm in the LP-DED one. However, the overall grain structure remained unchanged for both coupons after HT (see **Figure 6** (c)-(d)). The secondary phase particles, such as B_4C, TiC, Ti_5Si_3, and Al_3Ti acted as GB pinners that resisted

grain growth after HT. In addition, the pole figures for both processes in NHT and HIP+T6 conditions showed no strong crystallographic texture, indicating randomly oriented grains.

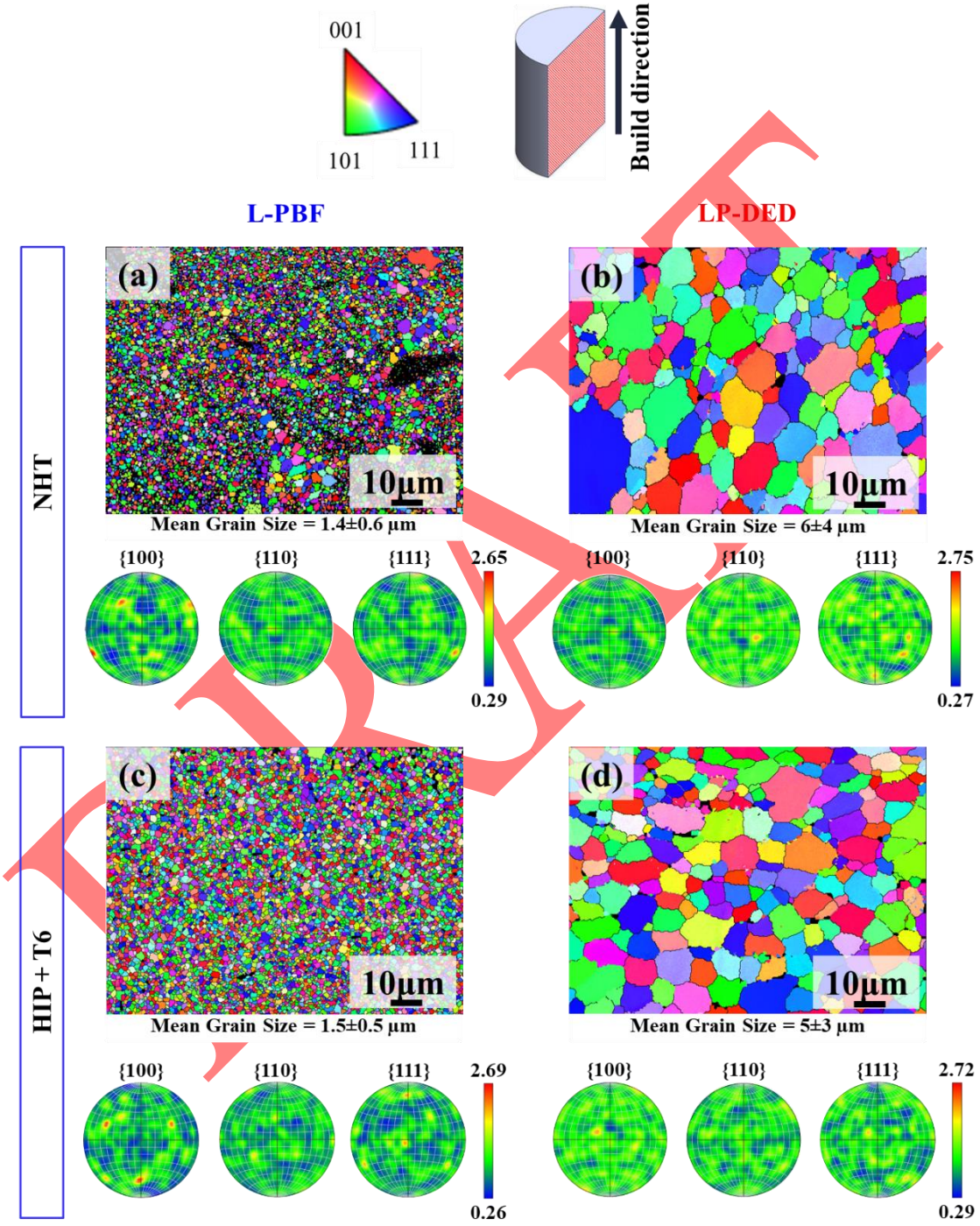


Figure 6. The IPF maps and pole figures from the EBSD analysis for both L-PBF and LP-DED A6061-RAM2 alloy in different heat treatment conditions: (a)-(b) NHT and (c)-(d) HIP+T6.

3.2. Room temperature tensile behavior

The room temperature engineering stress-engineering strain curves of L-PBF and LP-DED A6061-RAM2 specimens in the HIP+T6 condition are plotted in **Figure 7** (a), with one typical curve shown for each process. Dashed arrows on the stress-strain curves indicate the removal of the extensometer at 0.1 mm/mm strain, which is the limit of the extensometer range. Tensile properties, including YS, UTS, and EL, are presented in **Figure 7** (b) as bar charts for comparison between both processes. L-PBF specimens showed higher YS and UTS than LP-DED ones. The increased strength observe in L-PBF can be ascribed to three attributes, i.e., the presence of finer grains (by Hall–Petch strengthening mechanism) [57,58], more secondary phase particles, such as TiC and B_4C (by Orowan strengthening mechanism) [59,60], and more Mg_2Si precipitate (by precipitation strengthening mechanism) [61,62] compared to LP-DED. The yield point phenomenon, with an upper and a lower yield point, can be seen in the stress-strain curve of L-PBF specimens (see **Figure 7** (a)). This is likely due to the pinning and unpinning of the dislocations by the low amount of solid solution of Ti, Cu, and C, which creates a Cottrell atmosphere around the dislocations [63,64].

The L-PBF specimens show less significant strain hardening compared to the LP-DED ones, which is reflected by the smaller difference between YS and UTS. The low strain hardening capability can be attributed to the presence of finer grains in L-PBF which resulted in a higher density of grain boundaries. These boundaries act as strong sinks for dislocations, making the dislocation multiplication, a key mechanism for strain hardening, more difficult in L-PBF specimens [65,66].

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EL is comparable for both L-PBF and LP-DED specimens. Fractography images in **Figure 7** (c)-(f) reveal the room temperature fracture mechanisms for both processes. The fracture surfaces indicate ductile fracture behavior characterized by the presence of dimples. Although the dimples are larger in LP-DED (see **Figure 7** (f)) compared to L-PBF (see **Figure 7** (d)), the EL of ~20% was measured for both processes. These dimples form due to nucleation, growth, and coalescence of voids, which require plastic deformation. The higher density of secondary phase particles such as B_4C and TiC in L-PBF specimens promotes void nucleation. The presence of these particles has been confirmed by the EDS elemental maps of the fracture surfaces and presented in **Figure S2** in the Supplemental Material. The GB sliding can also facilitate void formation. However, the increased density of Mg_2Si particles along the GB for both processes can impede sliding (see **Figure 4** (g)-(h)). This impeded GB sliding contributes to the comparable EL observed for both processes.

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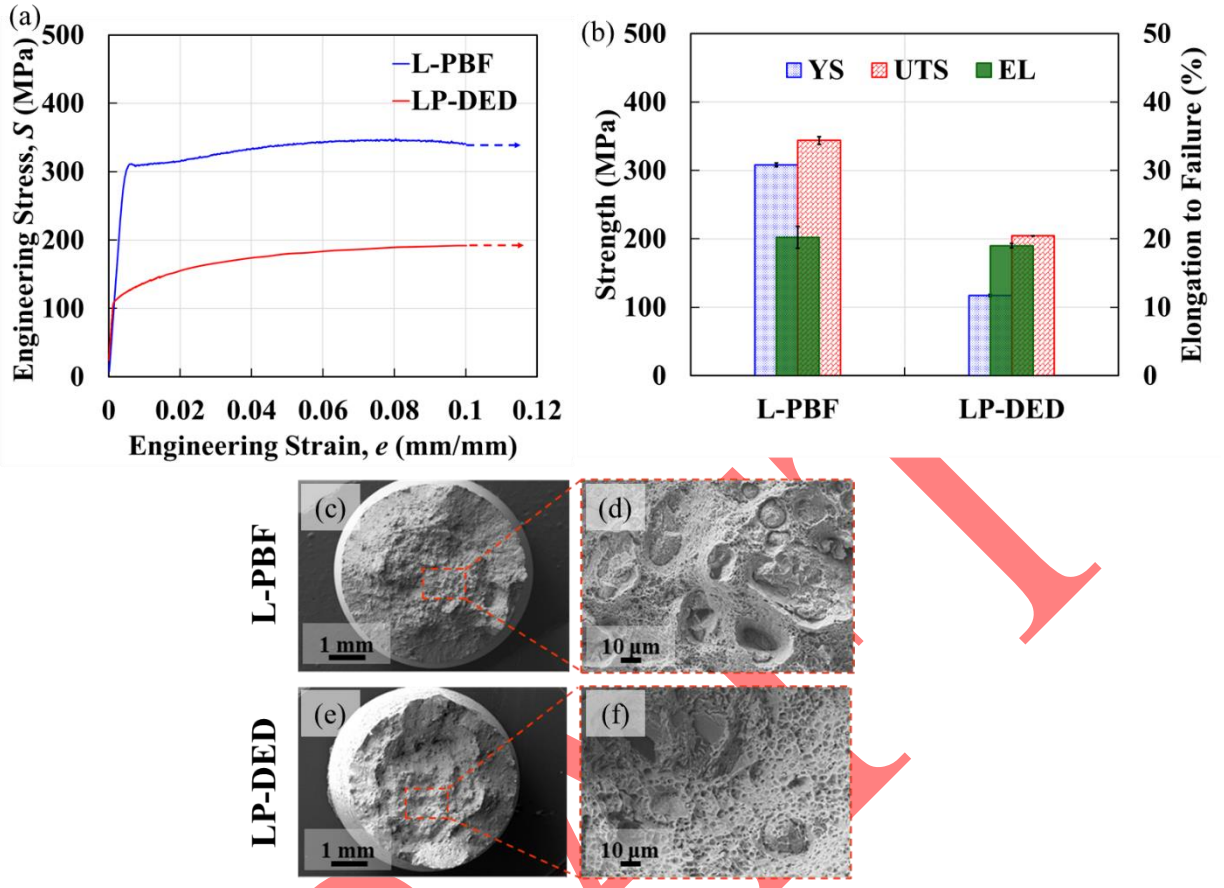


Figure 7. Comparison of the room temperature tensile behavior of L-PBF and LP-DED A6061-RAM2 specimens in HIP+T6 condition: (a) engineering stress-engineering strain curves, and (b) comparative bar charts for tensile properties. The dashed arrows on the stress-strain curves indicate that the extensometer was removed at approximately 0.1 mm/mm strain. Room temperature tensile fracture surfaces for (c)-(d) L-PBF and (e)-(f) LP-DED specimens.

3.3. Room temperature fatigue behavior

The strain-life fatigue test results at room temperature for L-PBF and LP-DED A6061-RAM2 specimens at strain amplitudes (ε_a) of 0.005 and 0.002 mm/mm are presented in **Figure 8**. At both ε_a, L-PBF specimens exhibited longer fatigue life than LP-DED ones.

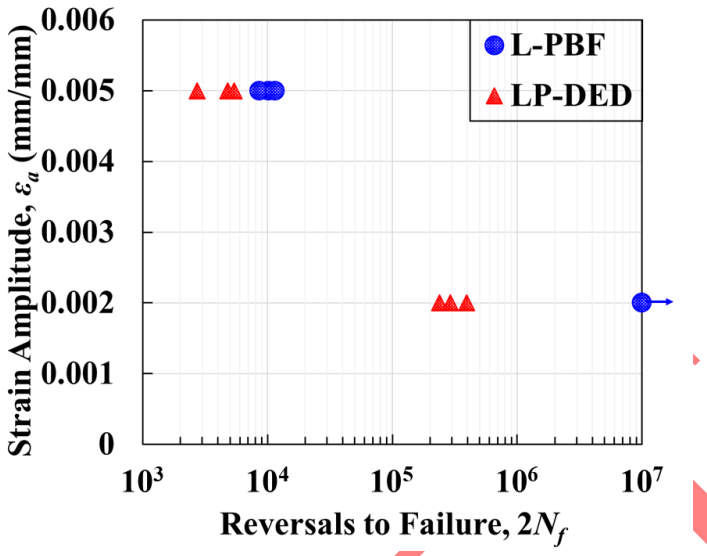


Figure 8. Comparison of the room temperature strain-life fatigue data between L-PBF and LP-DED A6061-RAM2 specimens.

Fractography of all failed specimens from both the L-PBF and LP-DED processes consistently reveals that cracks in L-PBF specimens initiate from volumetric defects, while cracks in LP-DED specimens originate from the fracture of secondary phase particles, specifically B_4C particles. In all fractured L-PBF specimens, the crack-initiating volumetric defects are larger than the B_4C particles present in the L-PBF specimens, whereas in all fractured LP-DED specimens, these crack-initiating particles are larger than the largest volumetric defects detected by the X-CT scan (see **Figure 3** and **Figure 4**). Representative fatigue fracture surfaces for both processes, including their initiation sites and crack growth regions, are shown in **Figure 9**. The EDS elemental maps of a representative fatigue crack initiation site of an LP-DED fractured specimen, presented in **Figure S3** of the Supplemental Material, confirm the presence of fractured B_4C as the crack initiation site.

At the higher ε_a of 0.005 mm/mm, the fatigue lives of specimens from both processes fell within the low-cycle fatigue regime, where fatigue life is primarily governed by crack growth rather than initiation. All the fracture surfaces from both processes at this ε_a reveal multiple crack

initiation sites and some representative fractographs of L-PBF and LP-DED specimens are illustrated in **Figure 9** (a)-(f). Fatigue striations, which formed during the ductile mode fatigue crack growth, are visible on the fracture surfaces. The spacing of these striations correlates with the incremental crack propagation per loading cycle, where larger spacing corresponds to faster crack growth. To ensure a valid comparison, regions with similar crack lengths on the fracture surfaces of specimens from two processes (highlighted by yellow dashed boxes in **Figure 9**) were analyzed. The striation spacing in the selected region of the L-PBF specimen (see **Figure 9** (c)) is finer than that of the LP-DED specimen (see **Figure 9** (f)), indicating a slower crack growth rate for the L-PBF specimen. This slower crack growth rate correlates with a longer fatigue life for L-PBF specimens compared to LP-DED ones.

At the low ε_a of 0.002 mm/mm, fatigue life is mainly influenced by crack initiation. Notably, at this ε_a, L-PBF specimens tested at room temperature exhibit runout (e.g., >10,000,000). However, fractography of all fractured LP-DED specimens reveals that fractured B_4C particles serve as crack initiation sites, and a representative fracture surface of an LP-DED specimen is presented in **Figure 9** (g)-(i)). Microstructure characterization showed that the size of these particles was larger in LP-DED specimens compared to L-PBF ones (see **Figure 4**). Particles of all sizes can act as stress concentrators; however, larger particles are more likely to initiate cracks due to their larger surrounding material volume experiencing the elevated stress compared to smaller particles [67–69]. The differences in the size of these crack-initiating particles contribute to the shorter fatigue life of LP-DED specimens tested at room temperature compared to L-PBF ones.

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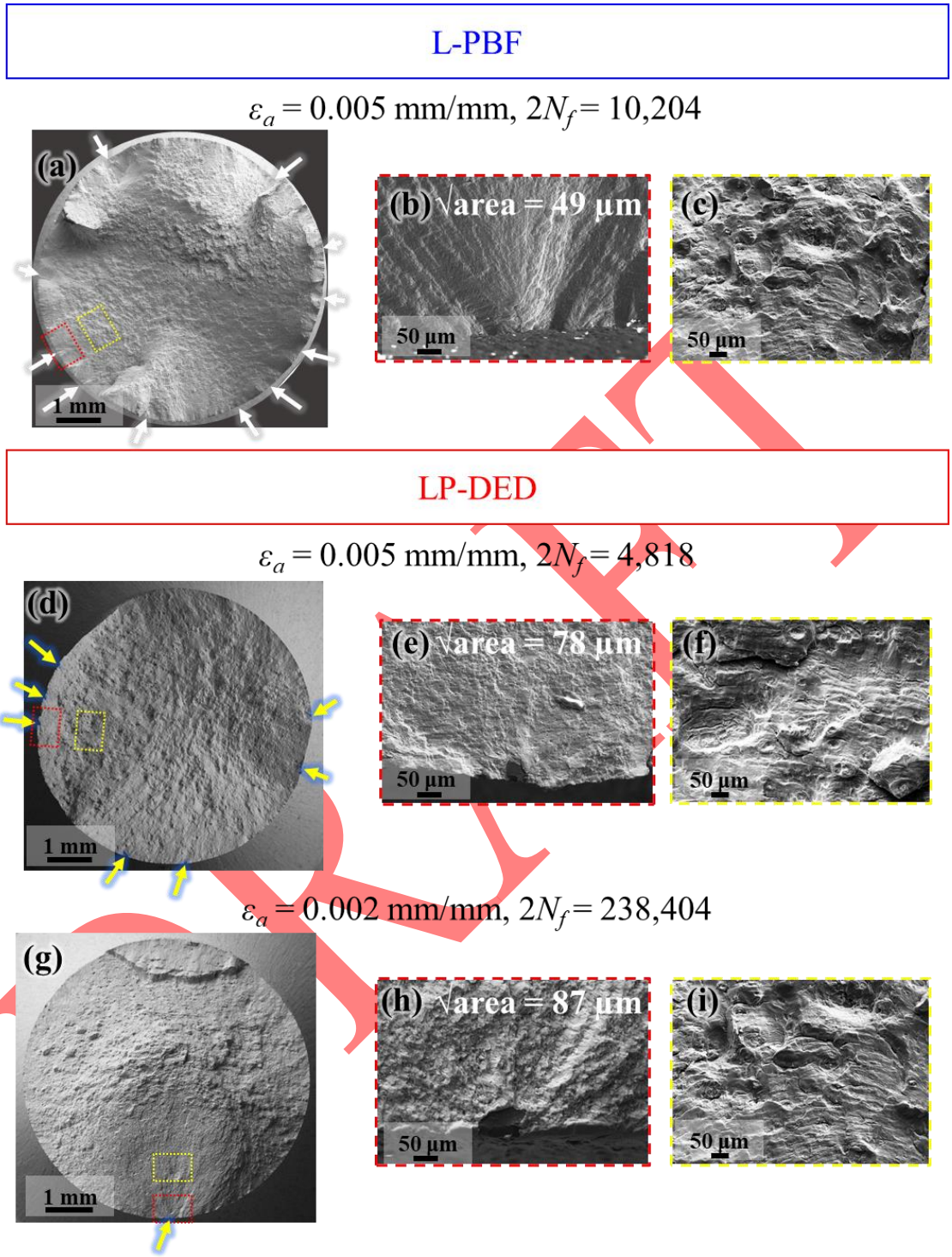


Figure 9. Fatigue fracture surfaces of A6061-RAM2 specimens for L-PBF and LP-DED tested at room temperature for (a)-(f) ε_a = 0.005 mm/mm and (g)-(i) ε_a = 0.002 mm/mm. White and yellow arrows indicate the crack initiation from volumetric defects and B_4C particles, respectively. Red and yellow dashed boxes indicate the crack initiation and crack growth regions, respectively.

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4. Effect of test temperatures

4.1. Tensile behavior

The tensile behavior of fully HT L-PBF and LP-DED A6061-RAM2 alloy was investigated across a temperature range from -195 °C up to 400 °C. Representative tensile engineering stress-engineering strain curves of L-PBF and LP-DED specimens at these test temperatures are plotted in **Figure 10** (a) and **Figure 10** (b), respectively (one typical curve is shown for each temperature. The bar charts comparing the uniaxial tensile properties (e.g., YS, UTS, and EL) of specimens from both processes at different temperatures are presented in **Figure 10** (c)-(d). The highest strength is observed at the cryogenic temperature (-195 °C), and the strength decreases with the rise of test temperatures for both processes. YS and UTS of both L-PBF and LP-DED specimens gradually decrease from -195 to 200 °C and drop dramatically after that. Notably, YS is higher for L-PBF specimens than for LP-DED ones at or below 200 °C; however, beyond that temperature, it is comparable. Additionally, the rate of decline in YS with the increasing test temperature is lower for LP-DED specimens compared to L-PBF ones. EL is mostly comparable between both the processes from -195 to 200 °C; after that, it increases more significantly for L-PBF specimens than for LP-DED ones.

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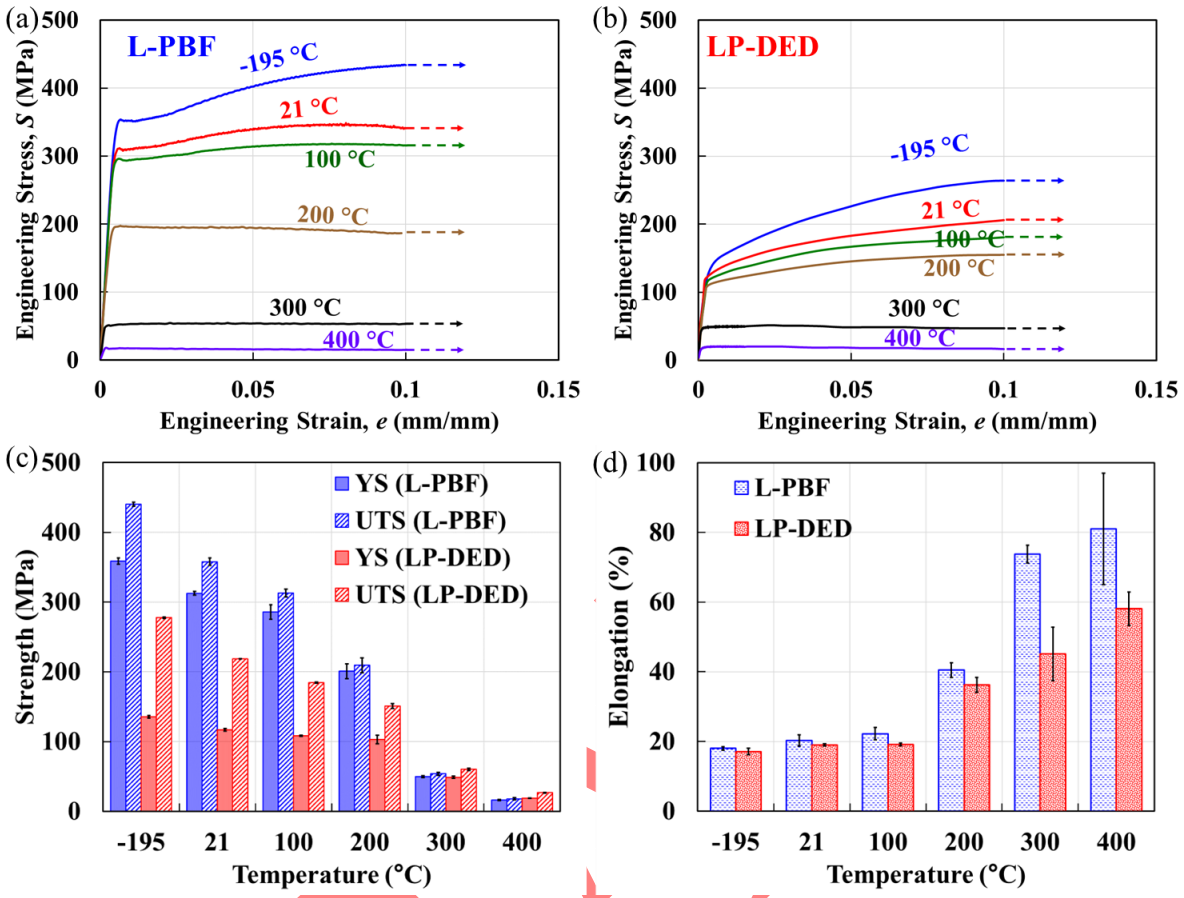


Figure 10. Tensile behavior of A6061-RAM2 alloy at fully heat-treated condition for different test temperatures: engineering stress-engineering strain curves for (a) L-PBF, and (b) LP-DED, and comparison of tensile properties of specimens for both processes, including (c) YS and UTS, (d) EL. The dashed arrows on the stress-strain curves indicate that the extensometer was removed at approximately 0.1 mm/mm strain, and the tests resumed under displacement control.

At -195 °C, specimens from both processes show the highest YS, which can be attributed to the low dislocation mobility at such a low temperature. As the test temperature increases from -195 to 200 °C, the alloy exhibits a gradual decrease in YS, which can be attributed to a gradual increase in dislocation mobility [70]. The higher YS observes in L-PBF specimens compared to LP-DED specimens at or below 200 °C due to their finer-grained microstructure, which inhibits dislocation motion more effectively compared to the coarser-grained microstructure in LP-DED

specimens. Above 200 °C, a drastic drop in YS is observed for both processes, indicating the activation of another deformation mechanism (e.g., excessive GB sliding). GB sliding is more pronounced in the finer grains of L-PBF specimens than in the coarser grains of LP-DED specimens.

UTS largely follows the same trend as YS for both processes. However, the differences between YS and UTS for L-PBF specimens are smaller compared to LP-DED ones, which can be attributed to the low strain hardening capability for L-PBF. The unpronounced strain hardening observed for L-PBF specimens due to their finer grain structure, which results in a high density of GB that act as dislocation sinks, limiting the interaction and multiplication of dislocations [71,72].

EL is similar between both processes up to 200 °C, with values remaining mostly unchanged up to 100 °C. Although the finer grains in L-PBF should favor the GB sliding and hence higher ductility compared to LP-DED, similar values are observed for both processes up to 200 °C. This can be attributed to the presence of stable Mg_2Si precipitates along the GB at these temperatures, which hinder excessive sliding [73]. This observation is consistent with the features observed on the microstructures (see **Figure 4** (k)-(l), and **Figure 5** (b) and (d)). The increased density of these particles in L-PBF prevents GB sliding more effectively, resulting in comparable EL for both processes up to 200 °C. Tensile fractography in **Figure 11** (a)-(l) indicates TiC and B_4C particles on fracture surfaces for specimens tested from -195 to 200 °C, which likely have nucleated the voids. These observations are similar to the room temperature tested tensile fracture surfaces (see **Figure 7** (c)-(f) and **Figure S2**). However, several studies have reported that above 200 °C (~50% of the melting temperature, i.e., $0.5T_m$) Mg_2Si starts to become incoherent and dissolves into the matrix for A6061 alloy [74–76]. The absence of these particles along the GB above 200 °C results in excessive GB sliding. At temperatures equal to or less than 200 °C, TiC and B_4C particles are

absent on the fracture surfaces (see **Figure 11** (m)-(t)), indicating GB sliding to enhance void nucleation and growth, which significantly increases ductility. The finer grains in L-PBF specimens facilitate higher GB sliding, promoting increased void formation to accommodate the applied deformation and contributing to the measure of higher EL compared to LP-DED specimens.

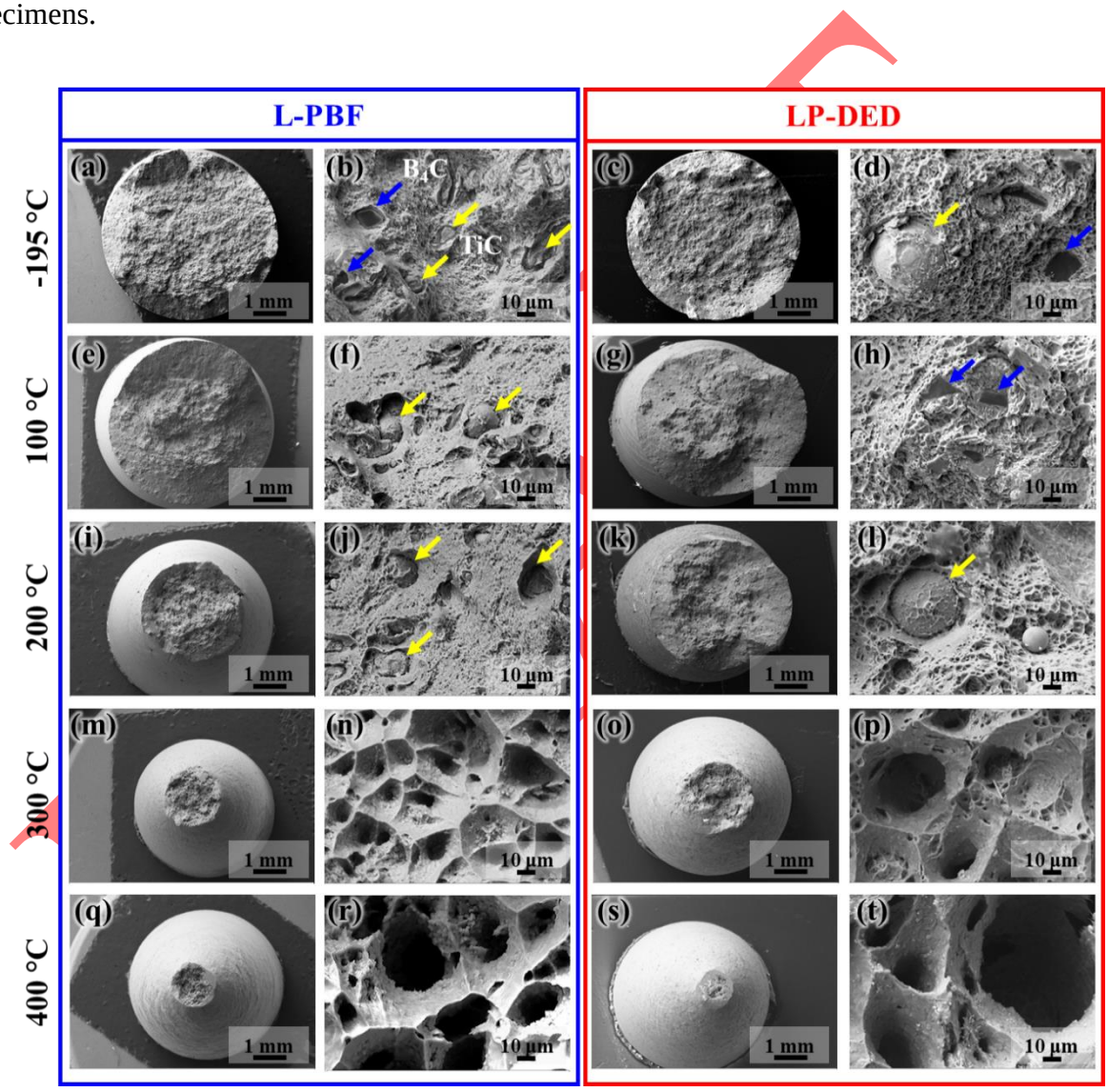


Figure 11. Tensile fracture surfaces of both L-PBF and LP-DED A6061-RAM2 alloy at different test temperatures: (a)-(d) -195 °C, (e)-(h) 100 °C, (i)-(l) 200 °C, (m)-(p) 300 °C, and (q)-(t) 400 °C. Blue and yellow arrows indicate the B_4C and TiC particles, respectively.

4.2. Fatigue behavior

The stable, mid-cycle stress-strain hysteresis loops from the fully reversed strain-controlled fatigue tests are shown in **Figure 12**. These tests were conducted at ε_a of 0.005 and 0.002 mm/mm, and at temperatures of -195, 21, 100, and 200 °C. The hysteresis loops for both L-PBF and LP-DED specimens exhibit significant plastic deformation at ε_a = 0.005 mm/mm for all test temperatures. As seen in **Figure 12** (a), (c), (e), and (g), the cyclic stress response in L-PBF specimens at ε_a = 0.005 mm/mm approaches or exceeds the YS of the alloy at those test temperatures (359, 313, 286, and 201 MPa at -195, 21, 100, and 200 °C, respectively) (see **Figure 10**). Additionally, L-PBF specimens exhibit narrower hysteresis loops, which indicate lower plasticity than LP-DED specimens. This variation can be attributed to the higher strength of L-PBF specimens as compared to LP-DED specimens. At ε_a = 0.002 mm/mm, specimens from both processes mostly show elastic deformation at -195, 21, and 100 °C, and some plasticity is observed at 200 °C.

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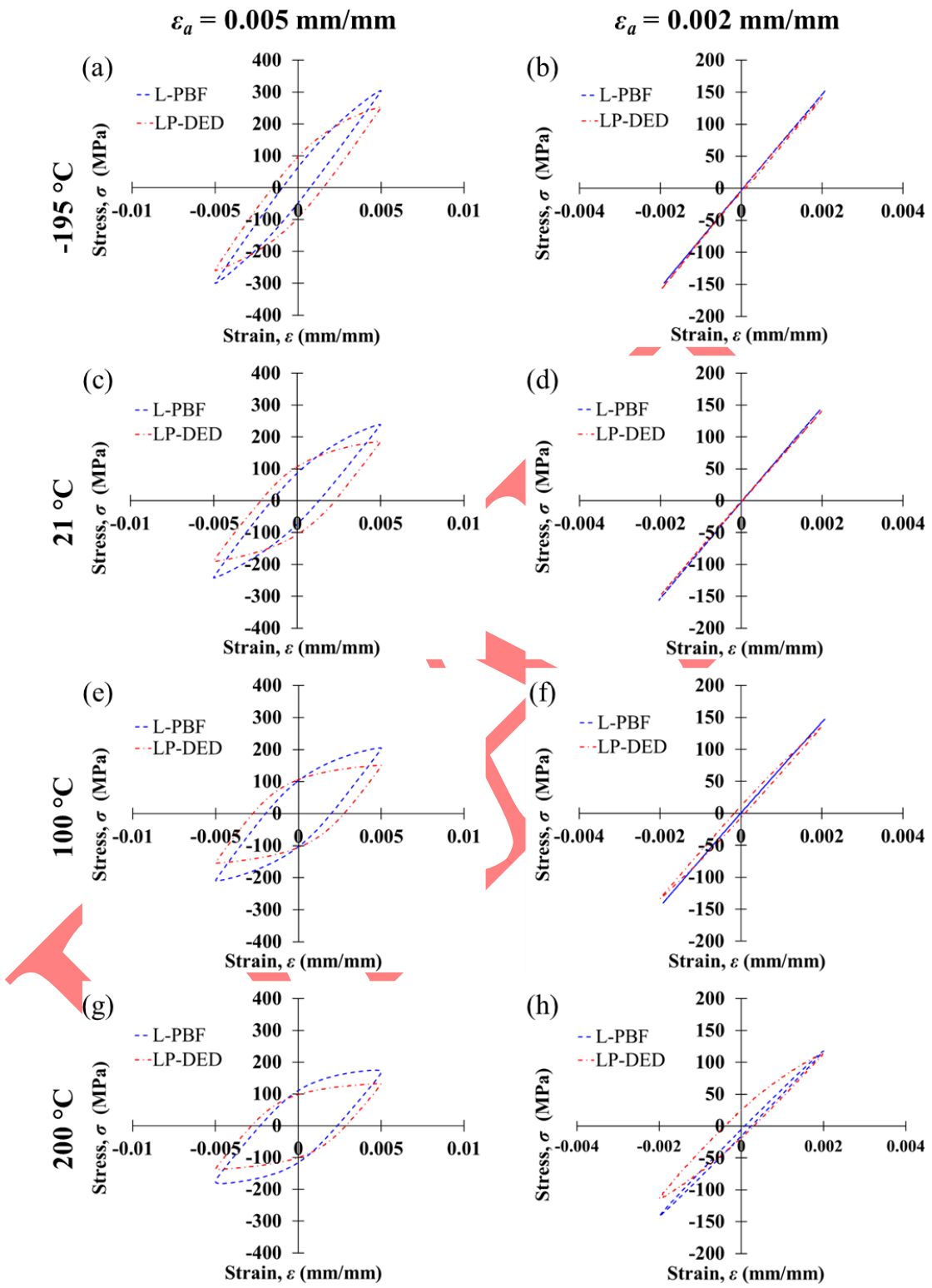


Figure 12. Stable, mid cycle hysteresis loops of uniaxial, fully reversed, strain-controlled fatigue tests for both L-PBF and LP-DED A6061-RAM2 specimens at different strain

amplitudes (ε_a): (a), (c), (e), (g) 0.005 mm/mm and (b), (d), (f), (h) 0.002 mm/mm at different test temperatures of -195, 21, 100, and 200 °C, respectively.

The fatigue data of L-PBF and LP-DED A6061-RAM2 specimens, including ε_a, stress amplitude (σ_a), mean stress (σ_m), and reversals to failure ($2N_f$), are listed in **Table 4**. The σ_a and σ_m values in **Table 4** are calculated at the stabilized cycle, i.e., mid-cycle, for each test. The fatigue life plot at strain amplitudes of 0.005 and 0.002 mm/mm for L-PBF and LP-DED A6061-RAM2 specimens tested at different temperatures are presented in **Figure 13**. With the exception of a few cases (i.e., ε_a = 0.005 mm/mm at 200 °C and ε_a = 0.002 mm/mm at -195 °C), L-PBF specimens exhibit superior fatigue performance compared to LP-DED specimens.

Table 4. Uniaxial, fully reversed, strain-controlled fatigue data of the L-PBF and LP-DED A6061-RAM2 specimens at different test temperatures.

Temperature (°C)	**ε_a (mm/mm)**		**σ_a (MPa)**		**σ_m (MPa)**		**$2N_f$ (Reversals)**	
	L-PBF	**LP-DED**	**L-PBF**	**LP-DED**	**L-PBF**	**LP-DED**	**L-PBF**	**LP-DED**
-195	0.005	0.005	302.45	257.39	2.41	-2.95	18,862	8,158
	0.005	0.005	299.46	261.78	2.74	-3.47	12,300	7,626
	0.005	0.005	320.07	259.85	-4.28	-1.97	16,156	5,728
	0.002	0.002	145.49	149.96	8.89	-8.66	>10,000,000	>10,000,000
	0.002	0.002	150.31	145.68	2.48	-4.28	>10,000,000	>10,000,000
	0.002	0.002	148.49	151.27	-1.59	-1.24	>10,000,000	>10,000,000
21	0.005	0.005	240.96	188.80	-1.54	-2.25	10,204	2,726
	0.005	0.005	239.75	205.00	-3.96	-2.78	8,574	4,818
	0.005	0.005	244.50	207.14	-1.42	-3.05	11,512	5,396
	0.002	0.002	149.49	146.94	-2.58	-3.01	>10,000,000	238,404
	0.002	0.002	151.97	144.50	-6.67	-3.78	>10,000,000	394,264
	0.002	0.002	153.61	148.19	-4.71	-2.47	>10,000,000	292,530
100	0.005	0.005	207.79	152.90	-2.28	-1.90	5,400	1,986
	0.005	0.005	251.44	143.57	-1.54	2.14	6,634	1,366
	0.005	0.005	257.65	144.69	-3.81	-4.72	7,598	2,442
	0.002	0.002	143.99	139.89	3.46	-3.54	584,542	63,544
	0.002	0.002	146.31	135.47	-4.57	-1.24	477,760	66,710
	0.002	0.002	143.80	137.51	-5.53	-2.78	536,786	131,048
200	0.005	0.005	181.45	124.14	-2.57	-2.25	1,006	1,984

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Temperature (°C)	ε_a (mm/mm)		σ_a (MPa)		σ_m (MPa)		$2N_f$ (Reversals)	
	L-PBF	LP-DED	L-PBF	LP-DED	L-PBF	LP-DED	L-PBF	LP-DED
	0.005	0.005	175.68	121.65	-1.91	-2.78	1,444	1,120
	0.005	0.005	171.67	128.47	-2.12	-3.05	1,938	1,088
	0.002	0.002	131.87	108.52	-1.51	-1.01	54,402	45,554
	0.002	0.002	127.52	101.84	-5.47	-1.71	133,610	38,240
	0.002	0.002	124.89	105.63	-2.71	-2.86	80,198	40,572

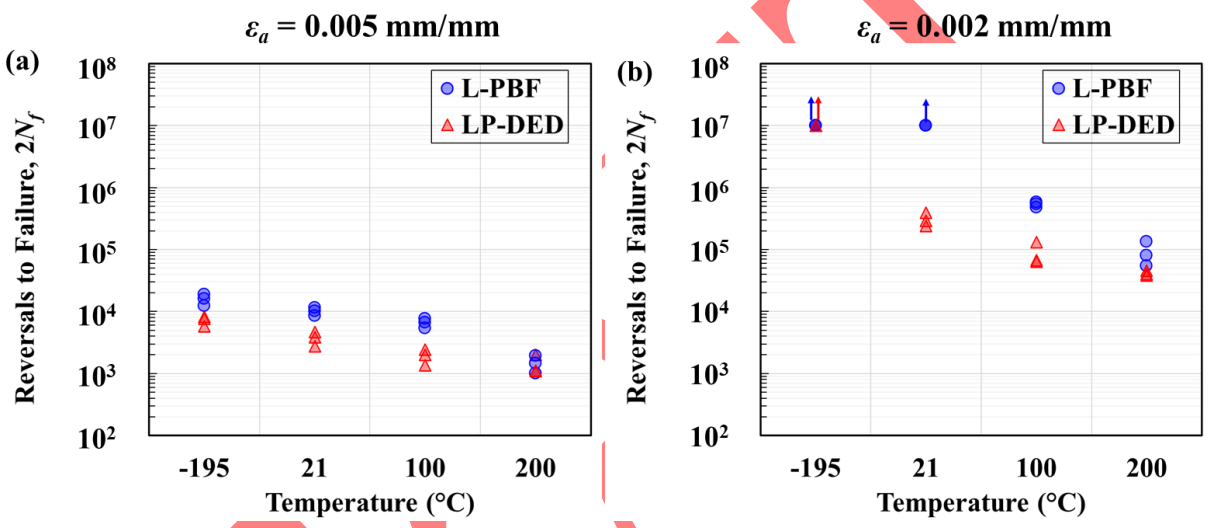


Figure 13. The fatigue life data at different test temperatures of L-PBF and LP-DED A6061-RAM2 specimens at strain amplitudes of (a) 0.005 mm/mm and (b) 0.002 mm/mm.

At a high ε_a of 0.005 mm/mm, fatigue life is mainly influenced by crack growth rather than initiation. As seen in **Figure 13** (a), fatigue life decreases with increasing test temperature as higher temperatures accelerate crack growth. L-PBF specimens tested at -195 and 100 °C under strain-controlled loading exhibit longer fatigue life compared to LP-DED ones, which can be attributed to the lower crack propagation rate in L-PBF specimens than that in LP-DED ones, and some representative fatigue fracture surfaces along with their crack initiation sites and crack growth regions are shown in **Figure 14**.

The fracture surfaces of L-PBF specimens tested at ε_a of 0.005 mm/mm and temperatures of -195 and 100 °C exhibited larger crack growth regions compared to those of LP-DED specimens (see **Figure 14** (a)-(d)). To compare the crack growth rate, fatigue striation spacing in these regions was analyzed, similar to the room temperature fatigue behavior at this ε_a. As shown in **Figure 14** (a)-(d), the selected crack growth regions for similar crack lengths, indicated by yellow dashed boxes, reveal that the striation spacing in L-PBF specimens is finer than that in LP-DED ones. This indicates a slower crack propagation rate in L-PBF specimens, which contributes to their longer fatigue lives compared to LP-DED ones at -195 and 100 °C.

However, at 200 °C, the crack growth rate is similar for both L-PBF and LP-DED specimens. This is evidenced by the comparable spacing of fatigue striations in the crack growth regions observed on the fracture surfaces (see **Figure 14** (e)-(f)). As a result, fatigue lives of the L-PBF and LP-DED specimens are comparable at this elevated temperature.

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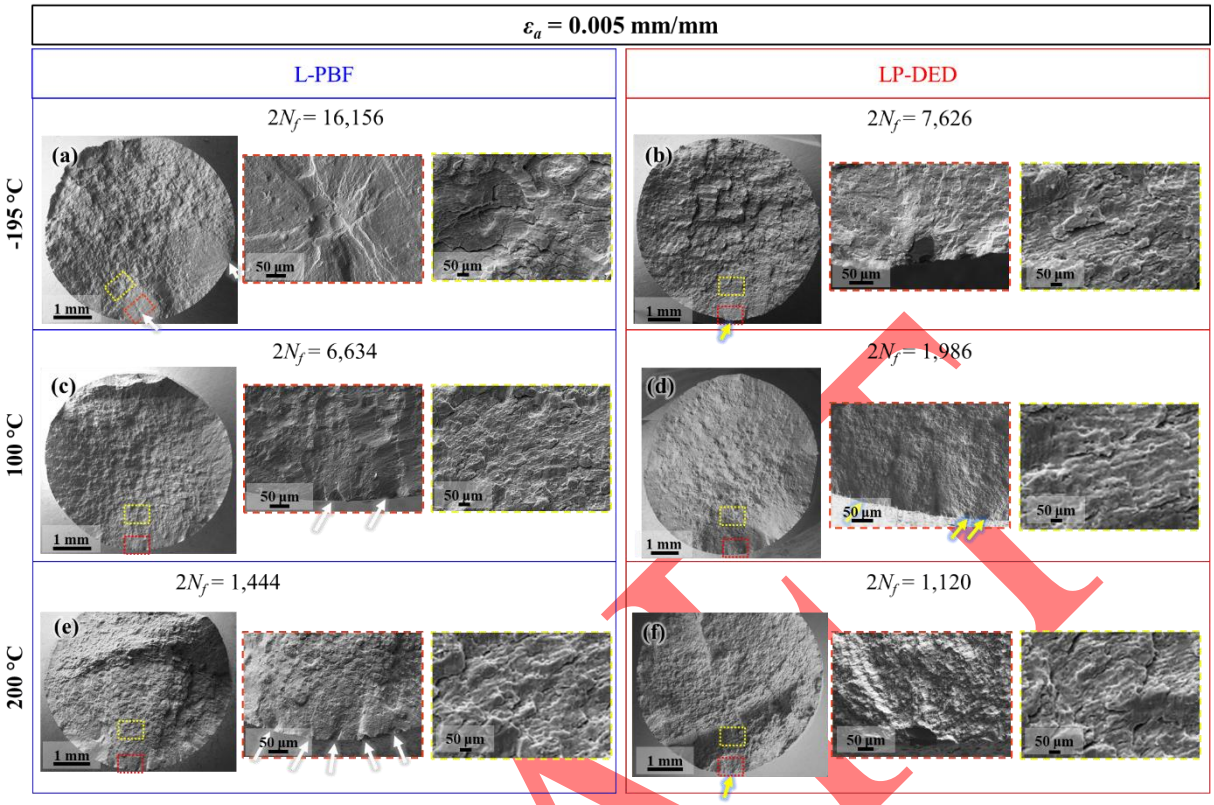


Figure 14. Fatigue fracture surfaces of specimens for L-PBF and LP-DED A6061-RAM2 alloy tested at the 0.005 mm/mm strain amplitude and at (a)-(b) -195 °C, (c)-(d) 100 °C, and (e)-(f) 200 °C White and yellow arrows indicate the volumetric defects and B_4C particles, respectively, as the crack-initiating sites. Red and yellow dashed boxes indicate the crack initiation and crack growth regions, respectively.

At the ε_a of 0.002 mm/mm, L-PBF specimens tested at 100 and 200 °C show increased fatigue life compared to LP-DED specimens, while fatigue life decreases with the increase in test temperature. At such a low ε_a, the fatigue life of a specimen is typically governed by crack initiation, and the shape and size of the fatigue-critical defects are of great importance for determining the fatigue life. Fractography was conducted on all the failed fatigue specimens for all test temperatures from both processes. The analysis shows that the fatigue cracks in the L-PBF specimens originate from volumetric defects, while in the LP-DED specimens, the cracks initiate from the fractured B_4C particles. Some of the representative fracture surfaces are shown in **Figure**

15 (a)-(d). To examine the influence of the size of defects and/or B_4C particles on fatigue lives, the reversals to failure are plotted against the measured crack-initiating defect √area (i.e., the square root of the area of the defect's projection onto the loading plane) for all the specimens from both processes tested at 100 and 200 °C at this ε_a and are presented in **Figure 16**. The analysis reveals that in LP-DED specimens, the crack-initiating B_4C particles are larger than the fatigue-critical volumetric defects in L-PBF specimens (see **Figure 15** and **Figure 16**). It leads to shorter fatigue life in the LP-DED specimens compared to those of L-PBF ones. For instance, an LP-DED specimen tested at 100 °C with a larger crack-initiating B_4C particle (√area = 89 μm) exhibits a shorter fatigue life ($2N_f$ = 63,544) than an L-PBF specimen with a smaller crack-initiating volumetric defect (√area = 41 μm) and longer fatigue life ($2N_f$ = 584,542). These results imply that the size of fatigue-critical defects and/or B_4C particles significantly affects fatigue life at such a low ε_a, alongside the effect of test temperature, highlighting a key difference in the fatigue behavior between the two processes.

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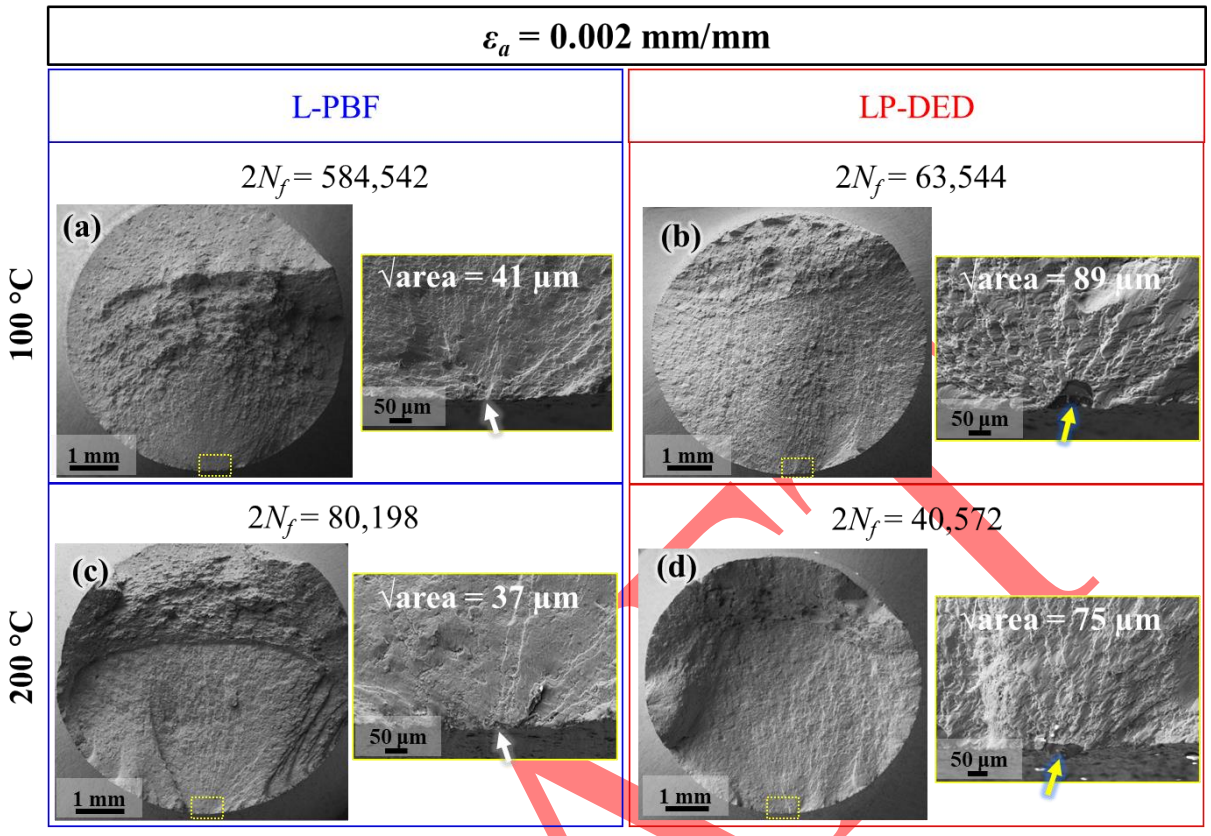


Figure 15. Fatigue fracture surfaces of specimens for L-PBF and LP-DED A6061-RAM2 alloy tested at the 0.002 mm/mm strain amplitude and at (a)-(b) 100 °C and (c)-(d) 200 °C. White and yellow arrows indicate the volumetric defects and B_4C particles, respectively, as the crack-initiating sites.

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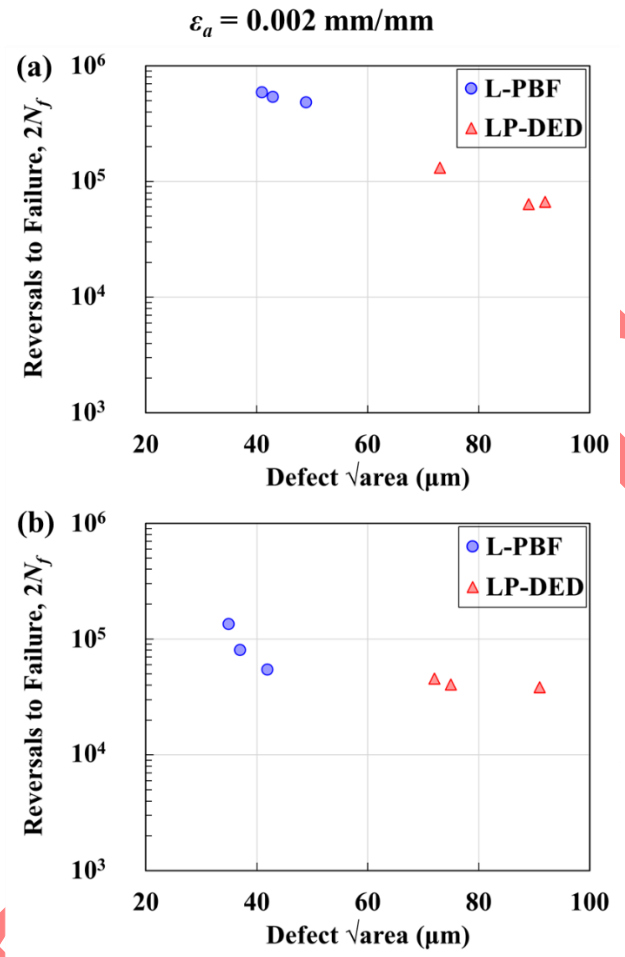


Figure 16. The reversals to failure vs. measured crack-initiating defect √area plots for L-PBF and LP-DED A6061-RAM2 specimens tested at ε_a of 0.002 mm/mm and temperatures of (a) 100 °C and (b) 200 °C. The blue and red markers indicate the √area of crack-initiating volumetric defects and B_4C particles, respectively.

5. Conclusion

The present study examined the temperature-dependent tensile and fatigue behaviors of A6061-RAM2 (2% TiC and B_4C inoculant nanoparticles) alloy fabricated via laser powder bed fusion (L-PBF) and laser powder direct energy deposition (LP-DED) processes. The effect of fabrication process and post process heat treatment (quenching media) on tensile and fatigue

behaviors across a wide temperature range was investigated. The following conclusions were drawn from this study:

1. Specimens from both processes showed a gradual decline in yield strength from -195 to 200 °C, due to the increased dislocation mobility. The finer L-PBF grain structures impeded dislocation motion more effectively and resulted in higher strengths compared to LP-DED specimens from -195 to 200 °C.
2. Elongation to failure (EL) was found to be similar for both processes at temperatures up to 200 °C, since the presence of Mg_2Si precipitates along the grain boundaries hindered grain boundary (GB) sliding, and a similar fracture mechanism was observed in both processes. Above 200 °C, the dissolution of those precipitates allowed for excessive GB sliding, which favored increased void formation to accommodate plastic deformation and significantly higher EL for fine-grained L-PBF specimens compared to coarse-grained LP-DED ones.
3. The fatigue performance of specimens from both processes at different test temperatures was dependent on crack initiation and their growth rate. At a high strain amplitude of 0.005 mm/mm, the L-PBF specimens showed longer fatigue lives than LP-DED specimens, tested from -195 to 100 °C, due to the finer microstructure, which favored slower crack propagation rates. The finer fatigue striation spacing observed in the crack growth regions of L-PBF specimens compared to LP-DED ones was consistent with this behavior.
4. At a low strain amplitude of 0.002 mm/mm, where fatigue life was mainly influenced by crack initiation, L-PBF specimens showed longer fatigue lives compared to LP-DED ones at both 100 and 200 °C. This was because in L-PBF specimens the fatigue cracks initiated from smaller volumetric defects, while in LP-DED specimens the initiation was from larger fatigue-critical B_4C particles.

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CRediT authorship contribution statement

Md Faysal Khan: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Vinay Kumar Yadav:** Formal analysis, Investigation, Methodology, Visualization. **Paul R. Gradl:** Conceptualization, Funding acquisition, Investigation, Methodology, Resources, Writing – review & editing. **Samad Firdosy:** Formal analysis, Investigation, Methodology, Visualization, Writing – review & editing. **Shuai Shao:** Conceptualization, Formal analysis, Investigation, Methodology, Resources, Writing – review & editing. **Nima Shamsaei:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Grabner F, Österreicher JA, Gruber B, Papenberg N, Gerstner F, Kirnstötter S, et al. Cryogenic Forming of Al-Mg Alloy Sheet for Car Outer Body Applications. Adv Eng Mater 2019;21:1900089. https://doi.org/10.1002/adem.201900089.

[2] Summers PT, Chen Y, Rippe CM, Allen B, Mouritz AP, Case SW, et al. Overview of aluminum alloy mechanical properties during and after fires. Fire Sci Rev 2015;4:3. https://doi.org/10.1186/s40038-015-0007-5.

[3] Afaf M. Abd El-Hameed, Y. A. Abdel-Aziz. Aluminium Alloys in Space Applications: A Short Report. J Adv Res Appl Sci Eng Technol 2021;22:1–7. https://doi.org/10.37934/araset.22.1.17.

[4] Madhloom MA, Ataiwi AH, Dawood JJ. Influence of cryogenic treatment on hardness, tensile properties, and microstructure of aluminum alloy AA6061. Mater Today Proc 2022;60:2157–61. https://doi.org/10.1016/j.matpr.2022.02.131.

[5] Rahman MA, Saleh T, Jahan MP, McGarry C, Chaudhari A, Huang R, et al. Review of Intelligence for Additive and Subtractive Manufacturing: Current Status and Future Prospects. Micromachines 2023, Vol 14, Page 508 2023;14:508. https://doi.org/10.3390/MI14030508.

[6] Martin JH, Yahata BD, Hundley JM, Mayer JA, Schaedler TA, Pollock TM. 3D printing of high-strength aluminium alloys. Nature 2017;549:365–9. https://doi.org/10.1038/nature23894.

[7] Uddin SZ, Murr LE, Terrazas CA, Morton P, Roberson DA, Wicker RB. Processing and characterization of crack-free aluminum 6061 using high-temperature heating in laser

powder bed fusion additive manufacturing. Addit Manuf 2018;22:405–15. https://doi.org/10.1016/j.addma.2018.05.047.

[8] Zhang X, Zheng H, Yu W. A review on solidification cracks in high-strength aluminum alloys via laser powder bed fusion. Mater Today Proc 2022;70:465–9. https://doi.org/10.1016/j.matpr.2022.09.366.

[9] Ding Y, Muñiz-Lerma JA, Trask M, Chou S, Walker A, Brochu M. Microstructure and mechanical property considerations in additive manufacturing of aluminum alloys. MRS Bull 2016;41:745–51. https://doi.org/10.1557/mrs.2016.214.

[10] Galy C, Le Guen E, Lacoste E, Arvieu C. Main defects observed in aluminum alloy parts produced by SLM: From causes to consequences. Addit Manuf 2018;22:165–75. https://doi.org/10.1016/j.addma.2018.05.005.

[11] Nuechterlein JS, Joseph Iten J. Reactive additive manufacturing. US10507638B2, 2021.

[12] Birol Y. Grain refining efficiency of Al–Ti–C alloys. J Alloys Compd 2006;422:128–31. https://doi.org/10.1016/j.jallcom.2005.11.059.

[13] Liu X, Liu Y, Huang D, Han Q, Wang X. Tailoring in-situ TiB2 particulates in aluminum matrix composites. Mater Sci Eng A 2017;705:55–61. https://doi.org/10.1016/j.msea.2017.08.047.

[14] Waller D, Polizzi A, Iten J. Feasibility Study of Additively Manufactured Al-6061 RAM2 Parts for Aerospace Applications. AIAA Scitech 2019 Forum, 2019. https://doi.org/10.2514/6.2019-0409.

[15] Gradl P, Tinker DC, Park A, Mireles OR, Garcia M, Wilkerson R, et al. Robust Metal

DRAFT

Additive Manufacturing Process Selection and Development for Aerospace Components. J Mater Eng Perform 2022;31:6013–44. https://doi.org/10.1007/s11665-022-06850-0.

[16] Zhang H, Zhu H, Qi T, Hu Z, Zeng X. Selective laser melting of high strength Al–Cu–Mg alloys: Processing, microstructure and mechanical properties. Mater Sci Eng A 2016;656:47–54. https://doi.org/10.1016/j.msea.2015.12.101.

[17] Khairallah SA, Anderson AT, Rubenchik A, King WE. Laser powder-bed fusion additive manufacturing: Physics of complex melt flow and formation mechanisms of pores, spatter, and denudation zones. Acta Mater 2016;108:36–45. https://doi.org/10.1016/j.actamat.2016.02.014.

[18] Thompson SM, Bian L, Shamsaei N, Yadollahi A. An overview of Direct Laser Deposition for additive manufacturing; Part I: Transport phenomena, modeling and diagnostics. Addit Manuf 2015;8:36–62. https://doi.org/10.1016/j.addma.2015.07.001.

[19] Piscopo G, Iuliano L. Current research and industrial application of laser powder directed energy deposition. Int J Adv Manuf Technol 2022;119:6893–917. https://doi.org/10.1007/s00170-021-08596-w.

[20] Hooper PA. Melt pool temperature and cooling rates in laser powder bed fusion. Addit Manuf 2018;22:548–59. https://doi.org/10.1016/j.addma.2018.05.032.

[21] Gibson I, Rosen D, Stucker B. Directed Energy Deposition Processes. Addit. Manuf. Technol., New York, NY: Springer New York; 2015, p. 245–68. https://doi.org/10.1007/978-1-4939-2113-3_10.

[22] Blakey-Milner B, Gradl P, Snedden G, Brooks M, Pitot J, Lopez E, et al. Metal additive

DRAFT

manufacturing in aerospace: A review. Mater Des 2021;209:110008. https://doi.org/10.1016/j.matdes.2021.110008.

[23] DebRoy T, Wei HL, Zuback JS, Mukherjee T, Elmer JW, Milewski JO, et al. Additive manufacturing of metallic components – Process, structure and properties. Prog Mater Sci 2018;92:112–224. https://doi.org/10.1016/j.pmatsci.2017.10.001.

[24] Nadammal N, Mishurova T, Fritsch T, Serrano-Munoz I, Kromm A, Haberland C, et al. Critical role of scan strategies on the development of microstructure, texture, and residual stresses during laser powder bed fusion additive manufacturing. Addit Manuf 2021;38:101792. https://doi.org/10.1016/j.addma.2020.101792.

[25] Oliveira JP, LaLonde AD, Ma J. Processing parameters in laser powder bed fusion metal additive manufacturing. Mater Des 2020;193:108762. https://doi.org/10.1016/j.matdes.2020.108762.

[26] Gradl PR, Cervone A, Gill E. Surface texture characterization for thin-wall NASA HR-1 Fe–Ni–Cr alloy using laser powder directed energy deposition (LP-DED). Adv Ind Manuf Eng 2022;4:100084. https://doi.org/10.1016/j.aime.2022.100084.

[27] Shamsaei N, Yadollahi A, Bian L, Thompson SM. An overview of Direct Laser Deposition for additive manufacturing; Part II: Mechanical behavior, process parameter optimization and control. Addit Manuf 2015;8:12–35. https://doi.org/10.1016/j.addma.2015.07.002.

[28] Wei S, Tasan CC. On the plastic deformation of a CoCrFeNiW-C alloy at elevated temperatures: Part II. Grain boundary sliding and damage mechanisms. Acta Mater 2023;252:118898. https://doi.org/10.1016/j.actamat.2023.118898.

DRAFT

[29] Otto F, Dlouhý A, Somsen C, Bei H, Eggeler G, George EP. The influences of temperature and microstructure on the tensile properties of a CoCrFeMnNi high-entropy alloy. Acta Mater 2013;61:5743–55. https://doi.org/10.1016/j.actamat.2013.06.018.

[30] Shou WB, Yi DQ, Liu HQ, Tang C, Shen FH, Wang B. Effect of grain size on the fatigue crack growth behavior of 2524-T3 aluminum alloy. Arch Civ Mech Eng 2016;16:304–12. https://doi.org/10.1016/j.acme.2016.01.004.

[31] Cheng L, Liang X, Bai J, Chen Q, Lemon J, To A. On utilizing topology optimization to design support structure to prevent residual stress induced build failure in laser powder bed metal additive manufacturing. Addit Manuf 2019;27:290–304. https://doi.org/10.1016/j.addma.2019.03.001.

[32] Yadollahi A, Shamsaei N. Additive manufacturing of fatigue resistant materials: Challenges and opportunities. Int J Fatigue 2017;98:14–31. https://doi.org/10.1016/j.ijfatigue.2017.01.001.

[33] Thompson SM, Bian L, Shamsaei N, Yadollahi A. An overview of Direct Laser Deposition for additive manufacturing; Part I: Transport phenomena, modeling and diagnostics. Addit Manuf 2015;8:36–62. https://doi.org/10.1016/j.addma.2015.07.001.

[34] Shamsaei N, Yadollahi A, Bian L, Thompson SM. An overview of Direct Laser Deposition for additive manufacturing; Part II: Mechanical behavior, process parameter optimization and control. Addit Manuf 2015;8:12–35. https://doi.org/10.1016/j.addma.2015.07.002.

[35] Chen Y, Kong W, Yuan C, Liu S, Cai Y, Wang Y, et al. The effects of temperature and stress on the high-cycle fatigue properties of a Ni-based wrought superalloy. Int J Fatigue 2023;172:107669. https://doi.org/10.1016/j.ijfatigue.2023.107669.

DRAFT

[36] Lee J, Kim K, Choi J, Kim JG, Kim S. Comparative study on fatigue crack propagation behavior of Ti–6Al–4V products made by DED (direct energy deposition) and L-PBF (laser-powder bed fusion) process. J Mater Res Technol 2023;23:4499–512. https://doi.org/10.1016/j.jmrt.2023.02.096.

[37] Thurston KVS, Gludovatz B, Hohenwarter A, Laplanche G, George EP, Ritchie RO. Effect of temperature on the fatigue-crack growth behavior of the high-entropy alloy CrMnFeCoNi. Intermetallics 2017;88:65–72. https://doi.org/10.1016/j.intermet.2017.05.009.

[38] Khan MF, Ghiaasiaan R, Gradl PG, Firdosy S, Shao S, Shamsaei N. Effect of Heat Treatment on the Micro-/Defect-Structure and Tensile Behavior of Additively Manufactured A6061-RAM2 Aluminum Alloy. 2024 Int. SOLID Free. Fabr. Symp., 2024, p. 685–98. https://doi.org/https://doi.org/10.26153/tsw/58106.

[39] Rosito M, Vanzetti M, Padovano E, Gili F, Sampieri R, Bondioli F, et al. Processability of A6061 Aluminum Alloy Using Laser Powder Bed Fusion by In Situ Synthesis of Grain Refiners. Metals (Basel) 2023;13. https://doi.org/10.3390/met13061128.

[40] Torbati-Sarraf H, Torbati-Sarraf SA, Chawla N, Poursaee A. A comparative study of corrosion behavior of an additively manufactured Al-6061 RAM2 with extruded Al-6061 T6. Corros Sci 2020;174:108838. https://doi.org/10.1016/j.corsci.2020.108838.

[41] Sullivan E, Polizzi A, Iten J, Nuechterlein J, Domack M, Liu S. Microstructural characterization and tensile behavior of reaction synthesis aluminum 6061 metal matrix composites produced via laser beam powder bed fusion and electron beam freeform fabrication. Int J Adv Manuf Technol 2022;121:2197–218. https://doi.org/10.1007/s00170-

022-09443-2.

[42] ASTM International. ASTM E1409-13 Standard Test Method for Determination of Oxygen and Nitrogen in Titanium and Titanium Alloys by Inert Gas Fusion. West Conshohocken, PA 19428-2959 USA 2021. https://doi.org/10.1520/E1409-13R21.

[43] ASTM International. ASTM E1019-18 Standard Test Methods for Determination of Carbon, Sulfur, Nitrogen, and Oxygen in Steel, Iron, Nickel, and Cobalt Alloys by Various Combustion and Inert Gas Fusion Techniques. West Conshohocken, PA 19428-2959 USA 2018. https://doi.org/10.1520/E1019-18.

[44] SAE Aerospace. AMS2770: Heat Treatment of Wrought Aluminum Alloy Parts. Aerosp Mater Specif AMS2770 2011;Rev. J.

[45] ASTM F3301-18a. Standard for Additive Manufacturing Post Processing Methods Standard Specification for Thermal Post-Processing Metal Parts Made Via Powder Bed Fusion. ASTM Stand 2018:3. https://doi.org/10.1520/F3301-18A.

[46] ASTM International. ASTM E3: Standard Guide for Preparation of Metallographic Specimens. West Conshohocken, PA 19428-2959 USA 2012;03.01:1–12. https://doi.org/10.1520/E0003-11R17.

[47] ASTM International. ASTM E8/E8M Standard test methods for tension testing of metallic materials. West Conshohocken, PA 19428-2959 USA 2013. https://doi.org/10.1520/E0008.

[48] ASTM E1450-16. Standard Test Method for Tension Testing of Structural Alloys in Liquid Helium. ASTM Stand 2016;01:1–9. https://doi.org/10.1520/E1450-16.

[49] ASTM International. Standard Test Methods for Elevated Temperature Tension Tests of

Metallic Materials. ASTM Stand 2020. https://doi.org/10.1520/E0021-20.

[50] ASTM International. ASTM E606/E606M Standard Test Method for Strain-Controlled Fatigue Testing. West Conshohocken, PA 19428-2959 USA 2012. https://doi.org/10.1520/E0606-04E01.Copyright.

[51] Weingarten C, Buchbinder D, Pirch N, Meiners W, Wissenbach K, Poprawe R. Formation and reduction of hydrogen porosity during selective laser melting of AlSi10Mg. J Mater Process Technol 2015;221:112–20. https://doi.org/10.1016/j.jmatprotec.2015.02.013.

[52] Xu R, Li R, Yuan T, Yu C, Zhu H. Towards the hydrogen pore in additively manufactured AlMgScZr alloy: Influencing factors, formation kinetics mechanism. J Mater Sci Technol 2024;199:125–44. https://doi.org/10.1016/j.jmst.2024.01.092.

[53] Gu J, Gao M, Yang S, Bai J, Ding J, Fang X. Pore formation and evolution in wire + arc additively manufactured 2319 Al alloy. Addit Manuf 2019;30:100900. https://doi.org/10.1016/j.addma.2019.100900.

[54] Mclean N, Bermingham MJ, Colegrove P, Sales A, Soro N, Ng CH, et al. Effect of Hot Isostatic Pressing and heat treatments on porosity of Wire Arc Additive Manufactured Al 2319. J Mater Process Technol 2022;310:117769. https://doi.org/10.1016/j.jmatprotec.2022.117769.

[55] Chu MG, Premkumar MK. Mechanism of TiC formation in Al/TiCin situ metal-matrix composites. Metall Trans A 1993;24:2803–5. https://doi.org/10.1007/BF02659503.

[56] Rajan TPD, Pillai RM, Pai BC. Reinforcement coatings and interfaces in aluminium metal matrix composites. J Mater Sci 1998;33:3491–503.

DRAFT

https://doi.org/10.1023/A:1004674822751.

[57] Pande CS, Cooper KP. Nanomechanics of Hall–Petch relationship in nanocrystalline materials. Prog Mater Sci 2009;54:689–706. https://doi.org/10.1016/j.pmatsci.2009.03.008.

[58] Yang C, Liu Z, Zheng Q, Cao Y, Dai X, Sun L, et al. Ultrasound assisted in-situ casting technique for synthesizing small-sized blocky Al3Ti particles reinforced A356 matrix composites with improved mechanical properties. J Alloys Compd 2018;747:580–90. https://doi.org/10.1016/j.jallcom.2018.02.010.

[59] Zhu AW, Starke EA. Strengthening effect of unshearable particles of finite size: a computer experimental study. Acta Mater 1999;47:3263–9. https://doi.org/10.1016/S1359-6454(99)00179-2.

[60] Behm N, Yang H, Shen J, Ma K, Kecskes LJ, Lavernia EJ, et al. Quasi-static and high-rate mechanical behavior of aluminum-based MMC reinforced with boron carbide of various length scales. Mater Sci Eng A 2016;650:305–16. https://doi.org/10.1016/j.msea.2015.10.064.

[61] Linardi E, Haddad R, Lanzani L. Stability Analysis of the Mg2Si Phase in AA 6061 Aluminum Alloy. Procedia Mater Sci 2012;1:550–7. https://doi.org/10.1016/j.mspro.2012.06.074.

[62] Kahlenberg R, Wojcik T, Falkinger G, Krejci AL, Milkereit B, Kozeschnik E. On the precipitation mechanisms of β-Mg2Si during continuous heating of AA6061. Acta Mater 2023;261:119345. https://doi.org/10.1016/j.actamat.2023.119345.

[63] Le GM, Godfrey A, Hansen N. Structure and strength of aluminum with sub-

DRAFT

micrometer/micrometer grain size prepared by spark plasma sintering. Mater Des 2013;49:360–7. https://doi.org/10.1016/j.matdes.2013.01.018.

[64] Hariprasad S, Sastry SML, Jerina KL. Deformation behavior of a rapidly solidified fine grained Al-8.5%Fe-1.2%V-1.7%Si alloy. Acta Mater 1996;44:383–9. https://doi.org/10.1016/1359-6454(95)00160-1.

[65] Kunz A, Pathak S, Greer JR. Size effects in Al nanopillars: Single crystalline vs. bicrystalline. Acta Mater 2011;59:4416–24. https://doi.org/10.1016/j.actamat.2011.03.065.

[66] Chisholm C, Bei H, Lowry MB, Oh J, Syed Asif SA, Warren OL, et al. Dislocation starvation and exhaustion hardening in Mo alloy nanofibers. Acta Mater 2012;60:2258–64. https://doi.org/10.1016/j.actamat.2011.12.027.

[67] Qian G, Lei W-S, Yu Z, Berto F. Statistical size scaling of breakage strength of irregularly-shaped particles. Theor Appl Fract Mech 2019;102:51–8. https://doi.org/10.1016/j.tafmec.2019.04.008.

[68] Hahn GT, Rosenfield AR. Metallurgical factors affecting fracture toughness of aluminum alloys. Metall Trans A 1975;6:653–68. https://doi.org/10.1007/BF02672285.

[69] KUMAI S, YOSHIDA K, HIGO Y, NUNOMURA S. Effects of the particle distribution on fatigue crack growth in particulate SiC/6061 aluminium alloy composites. Int J Fatigue 1992;14:105–12. https://doi.org/10.1016/0142-1123(92)90086-R.

[70] Khan MF, Ghiaasiaan R, Gradl PR, Shao S, Shamsaei N. Additively Manufactured Scalmalloy via Laser Powder Bed Fusion (L-PBF): Temperature-Dependent Tensile and Fatigue Behaviors. Fatigue Fract Eng Mater Struct 2025;n/a.

DRAFT

https://doi.org/10.1111/ffe.14549.

[71] Hu T, Jiang L, Yang H, Ma K, Topping TD, Yee J, et al. Stabilized plasticity in ultrahigh strength, submicron Al crystals. Acta Mater 2015;94:46–58. https://doi.org/10.1016/j.actamat.2015.04.044.

[72] Zhao L, Guo Q, Li Z, Li Z, Fan G, Xiong D-B, et al. Strain-rate dependent deformation mechanism of graphene-Al nanolaminated composites studied using micro-pillar compression. Int J Plast 2018;105:128–40. https://doi.org/10.1016/j.ijplas.2018.02.006.

[73] Hennum E, Marthinsen K, Tundal UH. Effect of Microstructure on the Precipitation of β-Mg2Si during Cooling after Homogenisation of Al-Mg-Si Alloys. Metals (Basel) 2024;14:215. https://doi.org/10.3390/met14020215.

[74] Doan LC, Ohmori Y, Nakai K. Precipitation and Dissolution Reactions in a 6061 Aluminum Alloy. Mater Trans JIM 2000;41:300–5. https://doi.org/10.2320/matertrans1989.41.300.

[75] Kreyca J, Kozeschnik E. Temperature-dependent strain hardening, precipitation and deformation-induced microstructure evolution in AA 6061. Mater Sci Eng A 2017;708:411–8. https://doi.org/10.1016/j.msea.2017.10.027.

[76] Kassner ME. Five-Power-Law Creep. In: Kassner ME, editor. Fundam. Creep Met. Alloy. Third Edit, Boston: Elsevier; 2015, p. 7–102. https://doi.org/10.1016/B978-0-08-099427-7.00002-5.

DRAFT